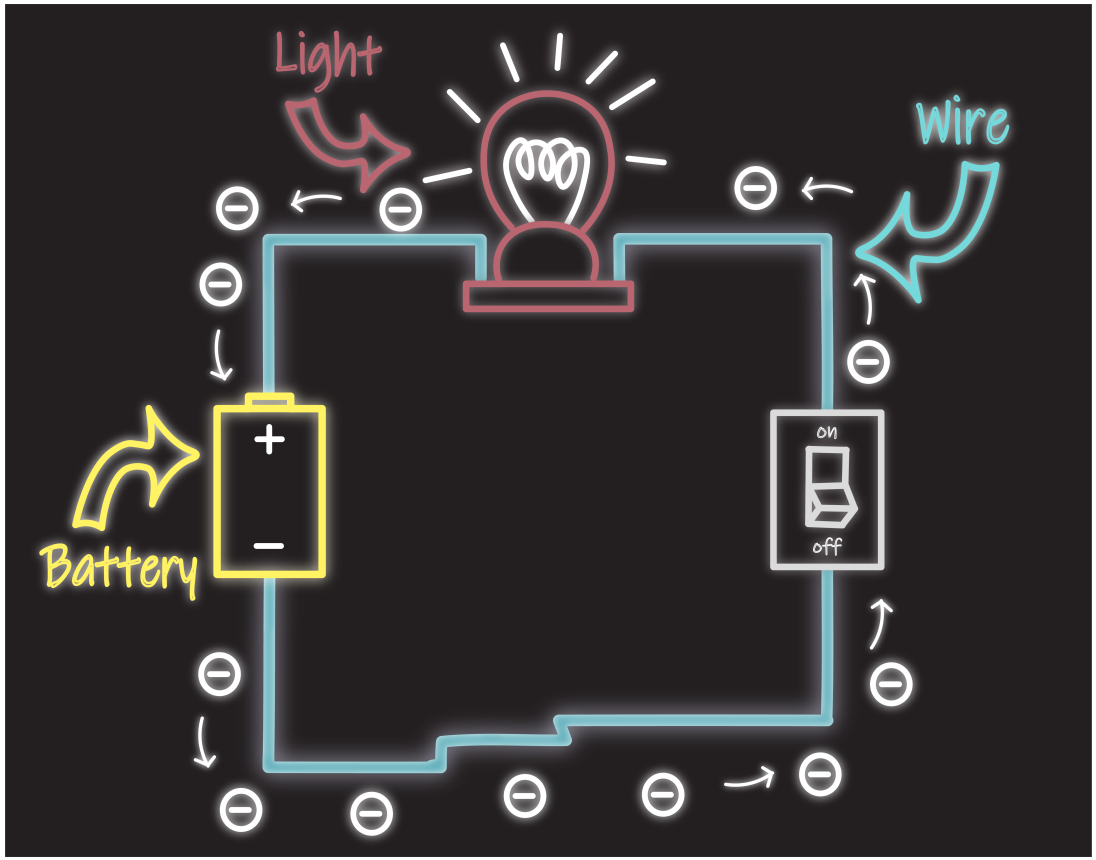
Quantum Electronics and Circuits

Brian Rashap

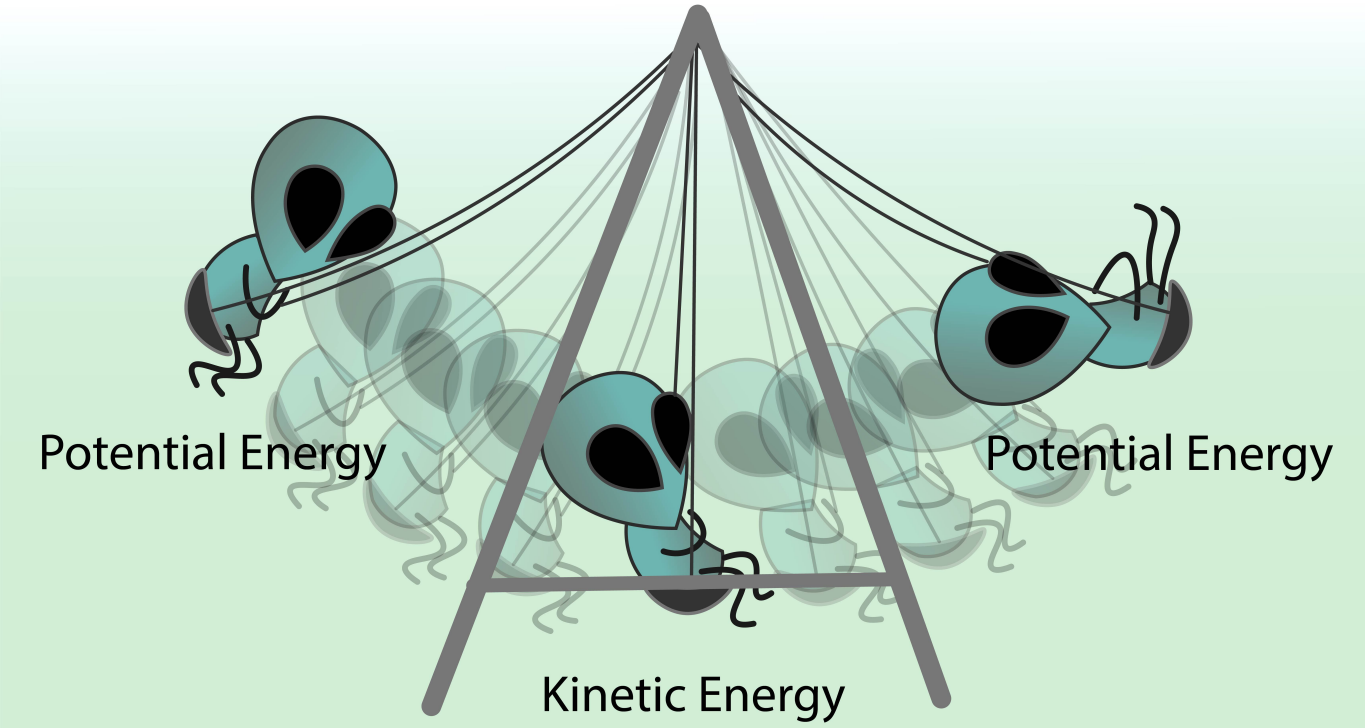
November 2025

Electrical Components and Circuits

Resistive Circuits



Energy



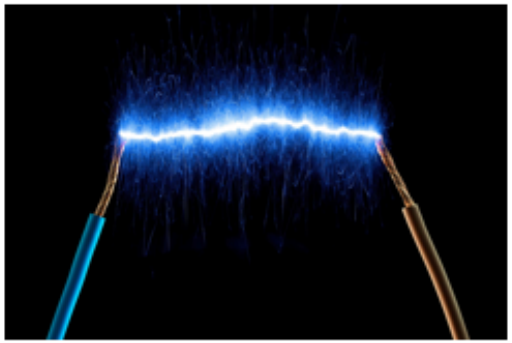
- Kinetic Energy - energy of motion
- Potential Energy - energy stored in an object

Electrical Circuit Terms

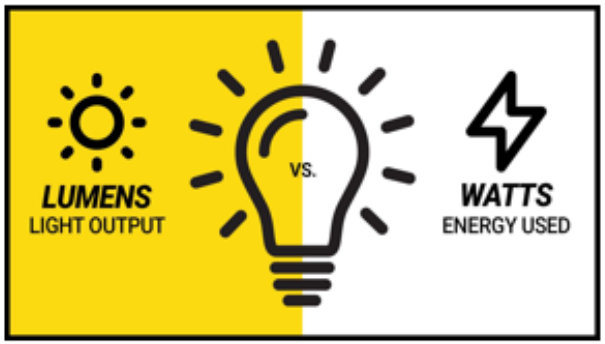


Voltage is **electric potential energy per unit charge** (V = J/C)

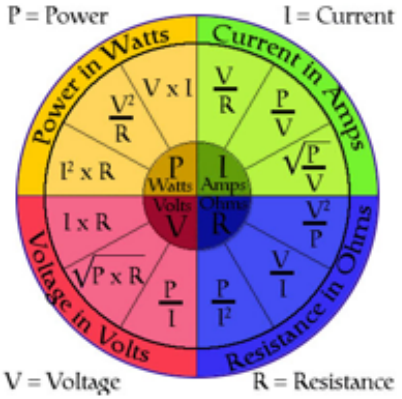
The Sub-atomic Particles			
Relative size	Name	Mass (Kg)	Charge (C)
	Proton	1.67×10^{-27}	$+1.602 \times 10^{-19}$
	Neutron	1.67×10^{-27}	0
* size exaggerated	Electron	9.11×10^{-31}	-1.602×10^{-19}



Electric current is the **rate of charge flow** (A = C/s)



Power is the rate of doing work or **the rate of using energy**. (W = J/S)

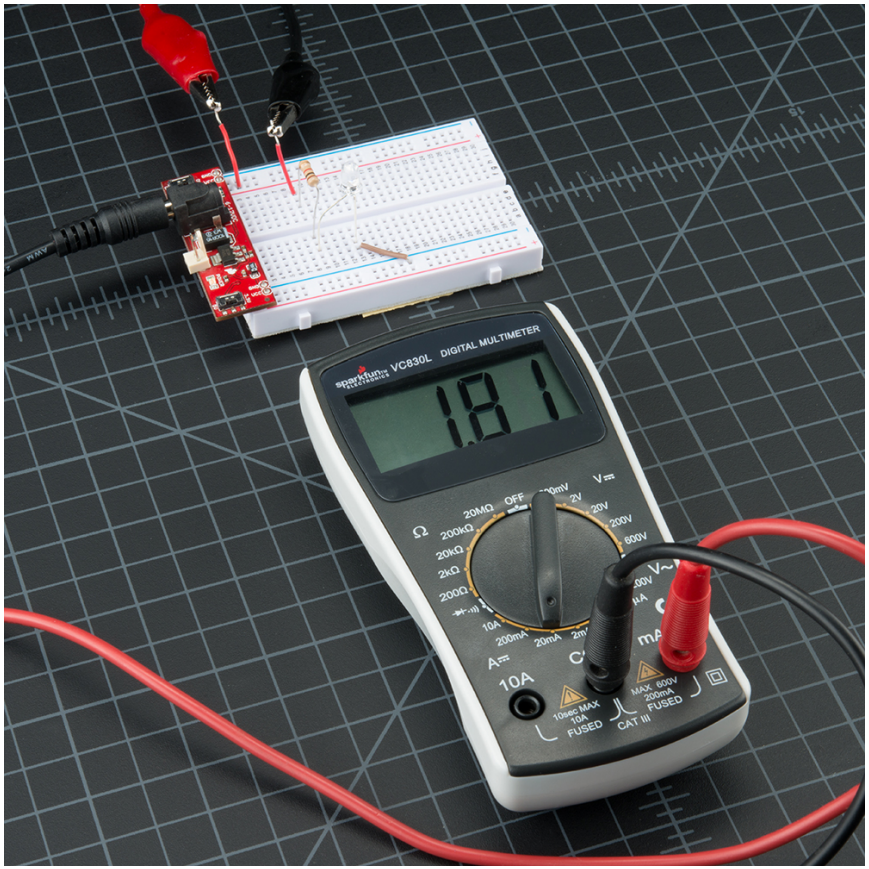


Power = Voltage x Current



Energy is the **amount of power** produced or consumed **over a given time**. J = W x s

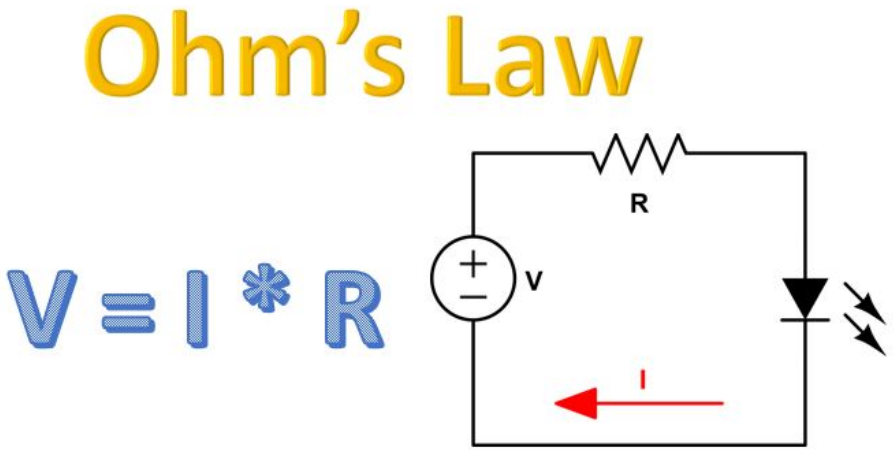
Measuring Voltage, Current, and Resistance

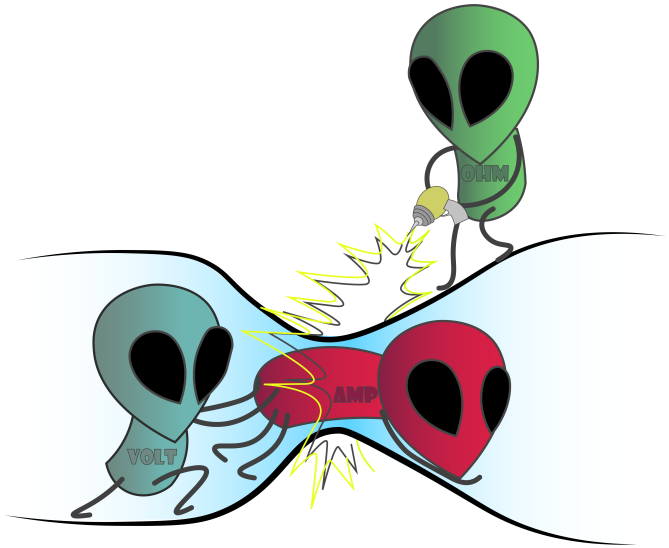




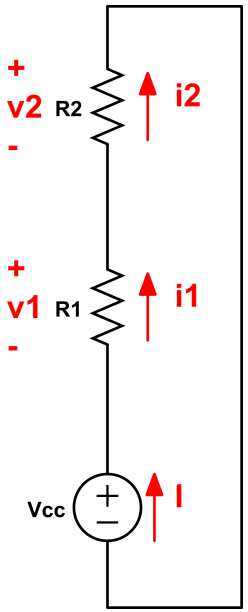
Ohm's Law

Georg Ohm (16 March 1789 – 6 July 1854) was a German physicist and mathematician. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist Alessandro Volta. Ohm found that there is a direct proportionality between the potential difference (voltage) applied across a conductor and the resultant electric current. This relationship is known as Ohm's law:





A word about circuit notation



Subscripts will be used to denote quanties (voltage, currect, etc) for different elements:

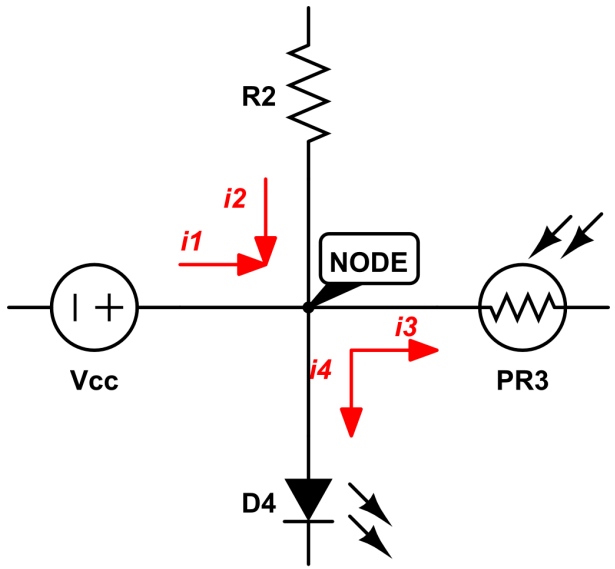
- i1 or i_1 is the current through Resistor 1 (R_1)
- i2 or i_2 is the current through Resistor 2 (R_2)
- v2 or v_2 is the votage across Resistor 2 (R_2)
- I is the current delivered by the power supply
- V_{cc} (common collector [a] voltage) is the notation will will use for $3.3V$ from the Particle

[a] Common Collector is a term for certain parts of transistor circuits. We will learn about Transistors in Lesson 11

Kirchhoff's First Law

Gustav Robert Kirchhoff (12 March 1824 – 17 October 1887) was a German physicist who contributed to the fundamental understanding of electrical circuits. His first law:

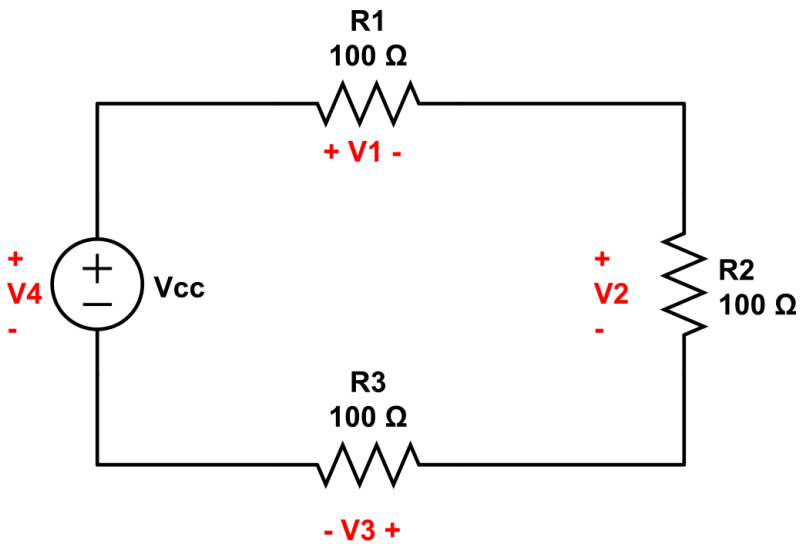
> In an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node



i1 + i2 = i3 + i4

Kirchhoff's Second Law

The directed sum of the potential differences (voltages) around any closed loop is zero.

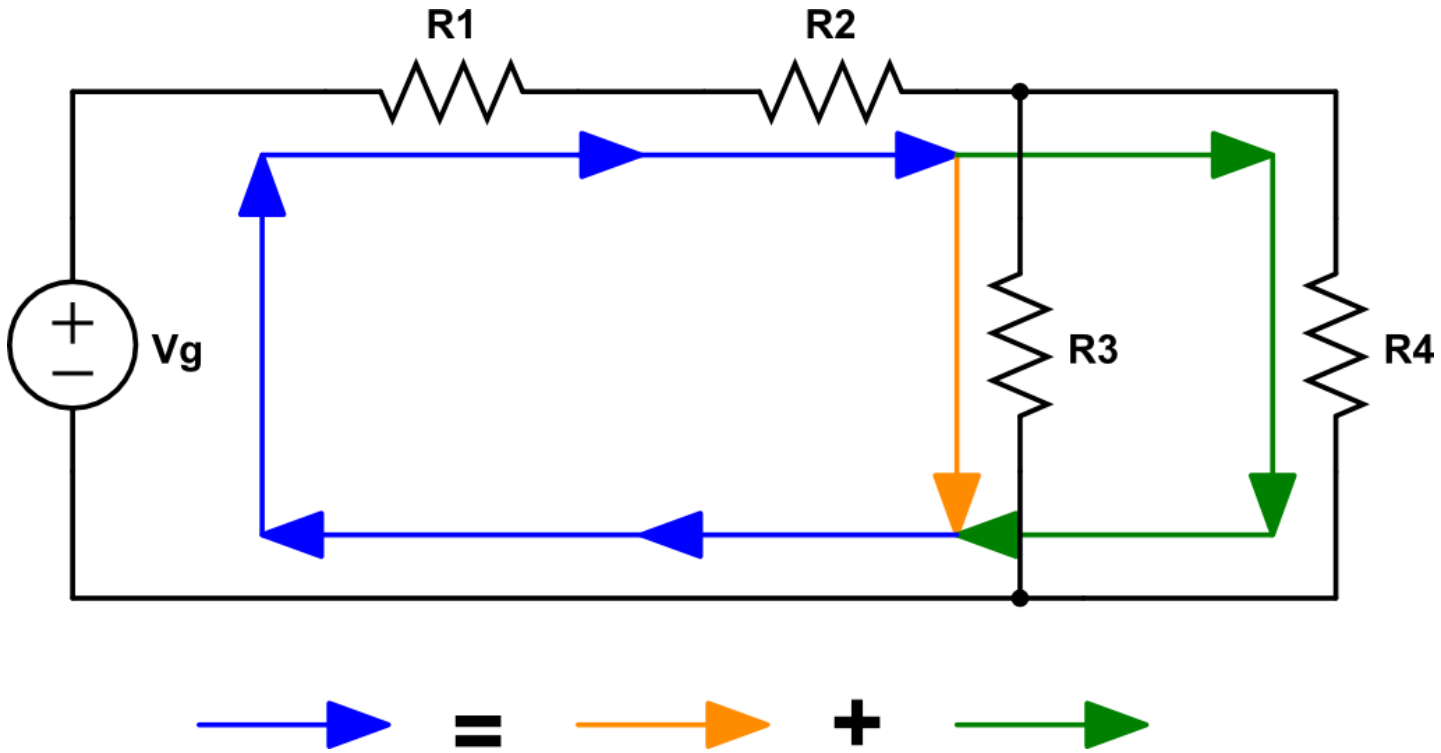


$$V4 - (V1 + V2 + V3) = 0$$

Kirchhoff's Second Law

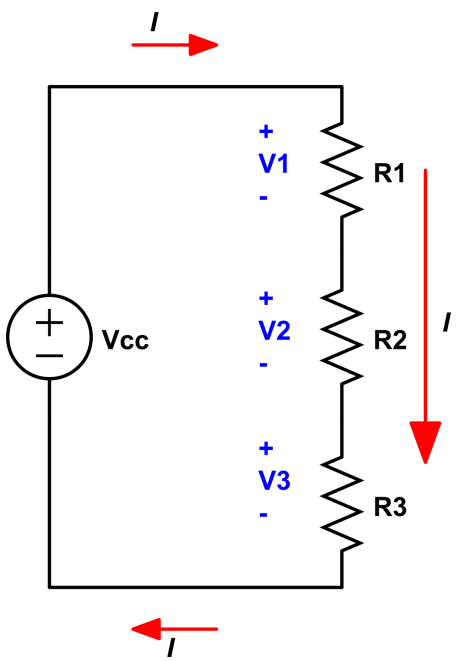


Resistors in Series and Parallel

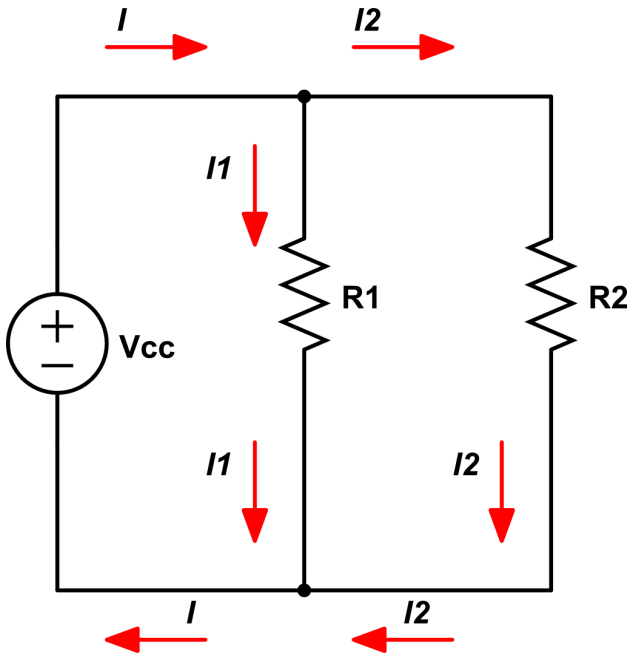


How many nodes? How many loops?

Resistors in Series and Parallel

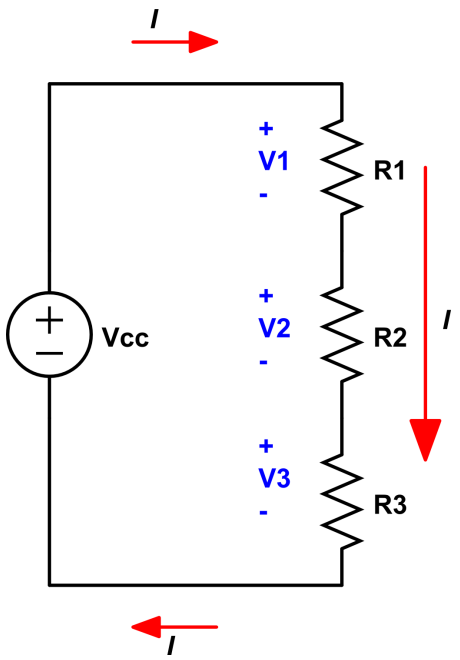


$$R_{eq} = R_1 + R_2 + R_3$$



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Resistors in Series



Node Law: $I = I_1 = I_2 = I_3$

Loop Law:

$$V_{cc} - (V_1 + V_2 + V_3) = 0$$

Rearranging the Loop Law:

$$V_{cc} = V_1 + V_2 + V_3 \quad (1)$$

Using Ohm's Law:

$$V_{cc} = IR_1 + IR_2 + IR_3 \quad (2)$$

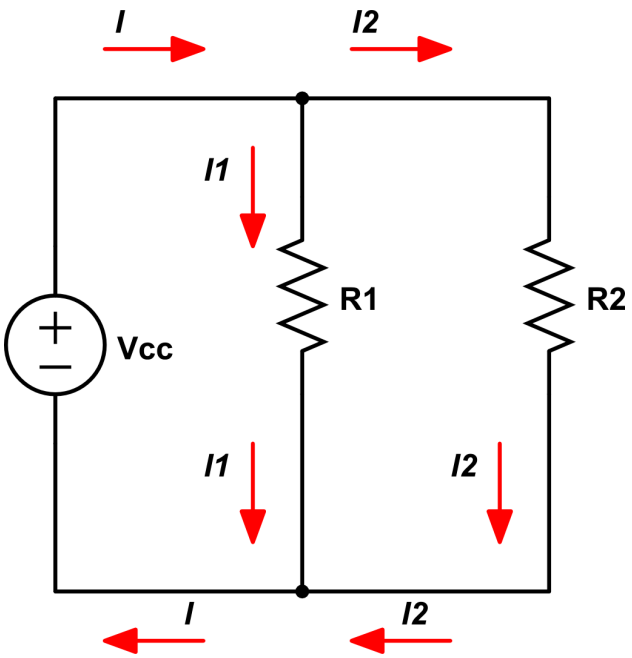
Using the Distributive Property:

$$V_{cc} = I(R_1 + R_2 + R_3) \quad (3)$$

Gives the Equivalent Resistance:

$$\boxed{R_{eq} = R_1 + R_2 + R_3} \quad (4)$$

Resistors in Parallel



Node Law: $I = I_1 + I_2$

Loop Law: $V_{cc} = V_1 = V_2$

$$I = I_1 + I_2 \tag{5}$$

$$I = \frac{V_1}{R_1} + \frac{V_2}{R_2} \tag{6}$$

$$I = \frac{V_{cc}}{R_1} + \frac{V_{cc}}{R_2} \tag{7}$$

$$\frac{V_{cc}}{R_{eq}} = V_{cc}(\frac{1}{R_1} + \frac{1}{R_2}) \tag{8}$$

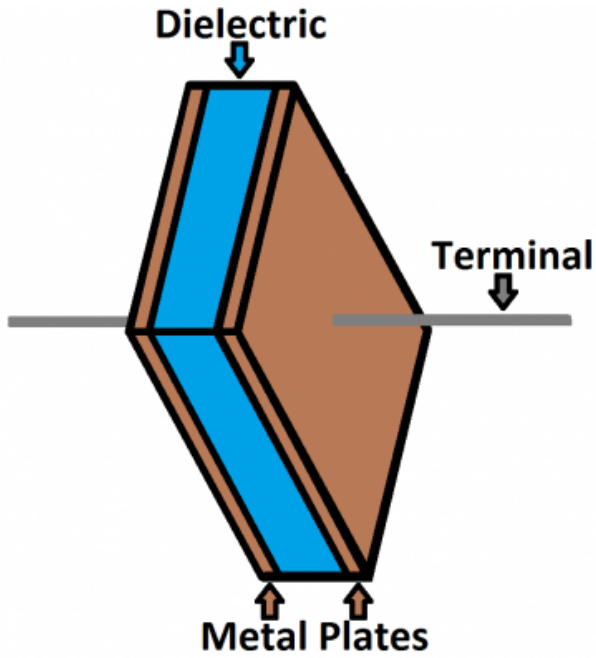
$$\boxed{\frac{1}{R_{eq}} = (\frac{1}{R_1} + \frac{1}{R_2})} \tag{9}$$

Capacitance and Filters

Capacitors

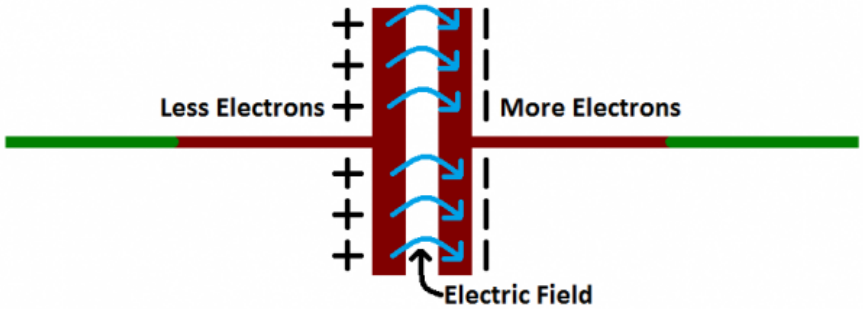
A capacitor is created out of two metal plates and an insulating material called a dielectric. The metal plates are placed very close to each other, in parallel, but the dielectric sits between them to make sure they don't touch.

- The dielectric can be made out of all sorts of insulating materials; paper, glass, rubber, ceramic, plastic, or anything that will impede the flow of current.
- The plates are made of a conductive material; aluminum, tantalum, silver, or other metals.



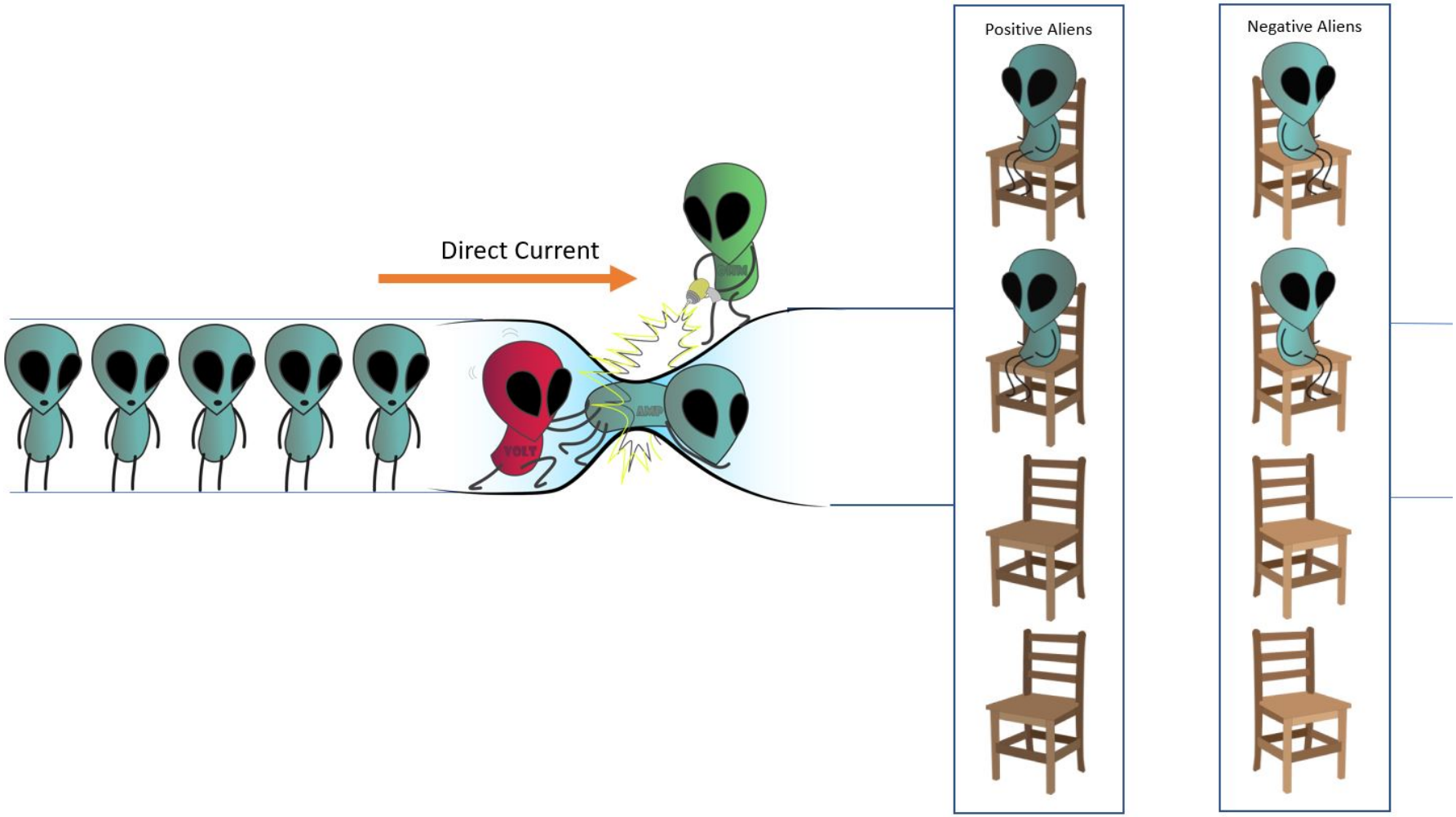
Capacitors

When current flows into a capacitor, the charges get "stuck" on the plates because they cannot get past the insulating dielectric. Electrons build up on one of the plates, and it becomes overall negatively charged. The large amount of negative charges pushes away like charges on the other plate, making it positively charged.

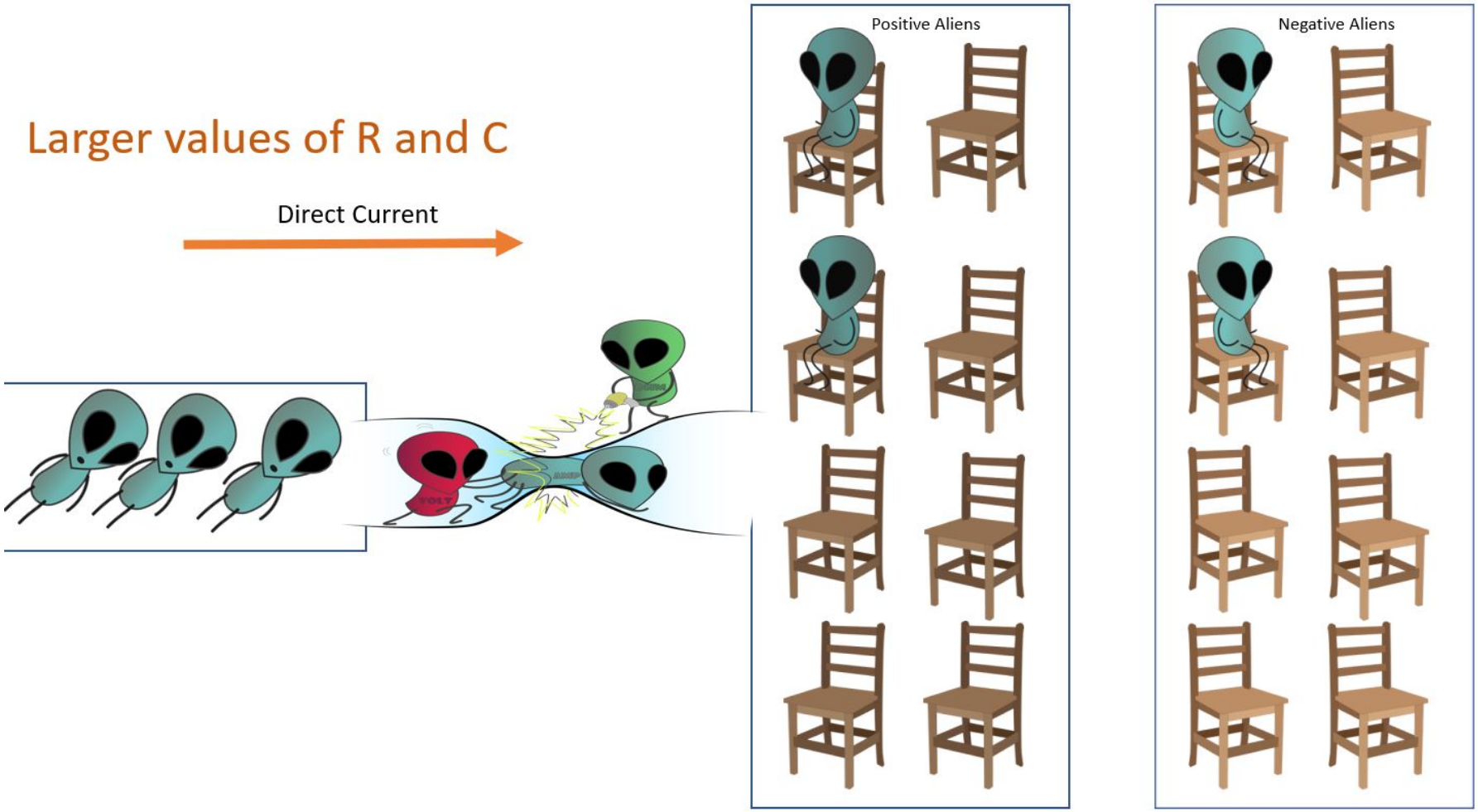


The stationary charges on these plates create an electric field, which influences electric potential energy and voltage. When charges group together on a capacitor like this, the capacitor is storing electric energy just as a battery might store chemical energy.

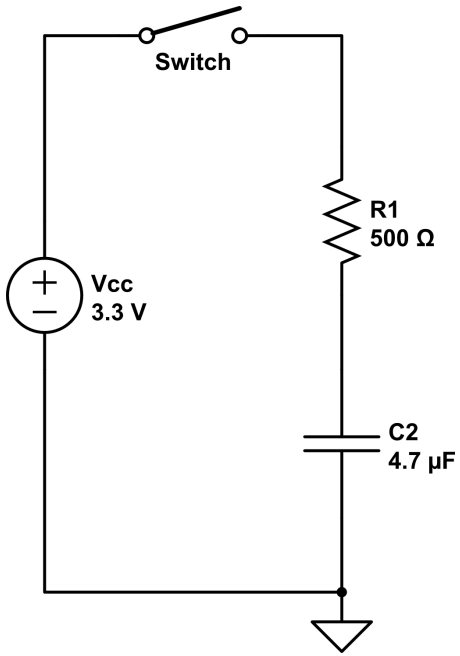
Capacitors in Circuits



Capacitors in Circuits



RC Time Constant



$$V_c(t) = V_c(0) * e^{\frac{t}{-\tau}}$$

Capacitance is defined as:

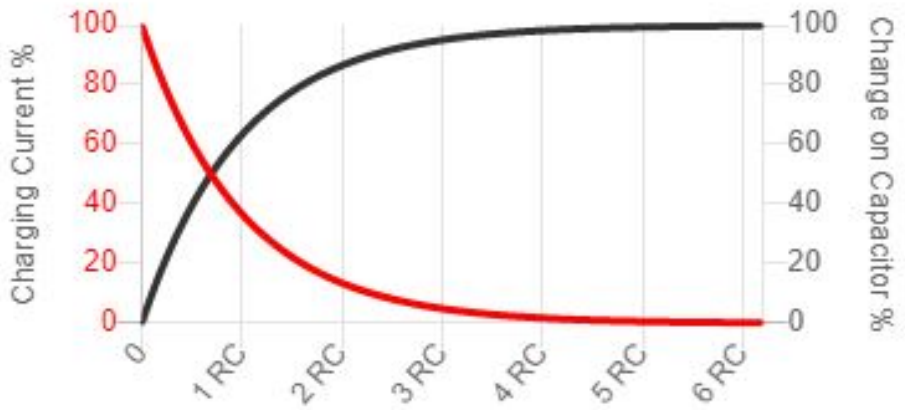
$$C = \frac{Q}{V}\left(\frac{Coulombs}{Volt}\right)$$

The current through a capacitor is:

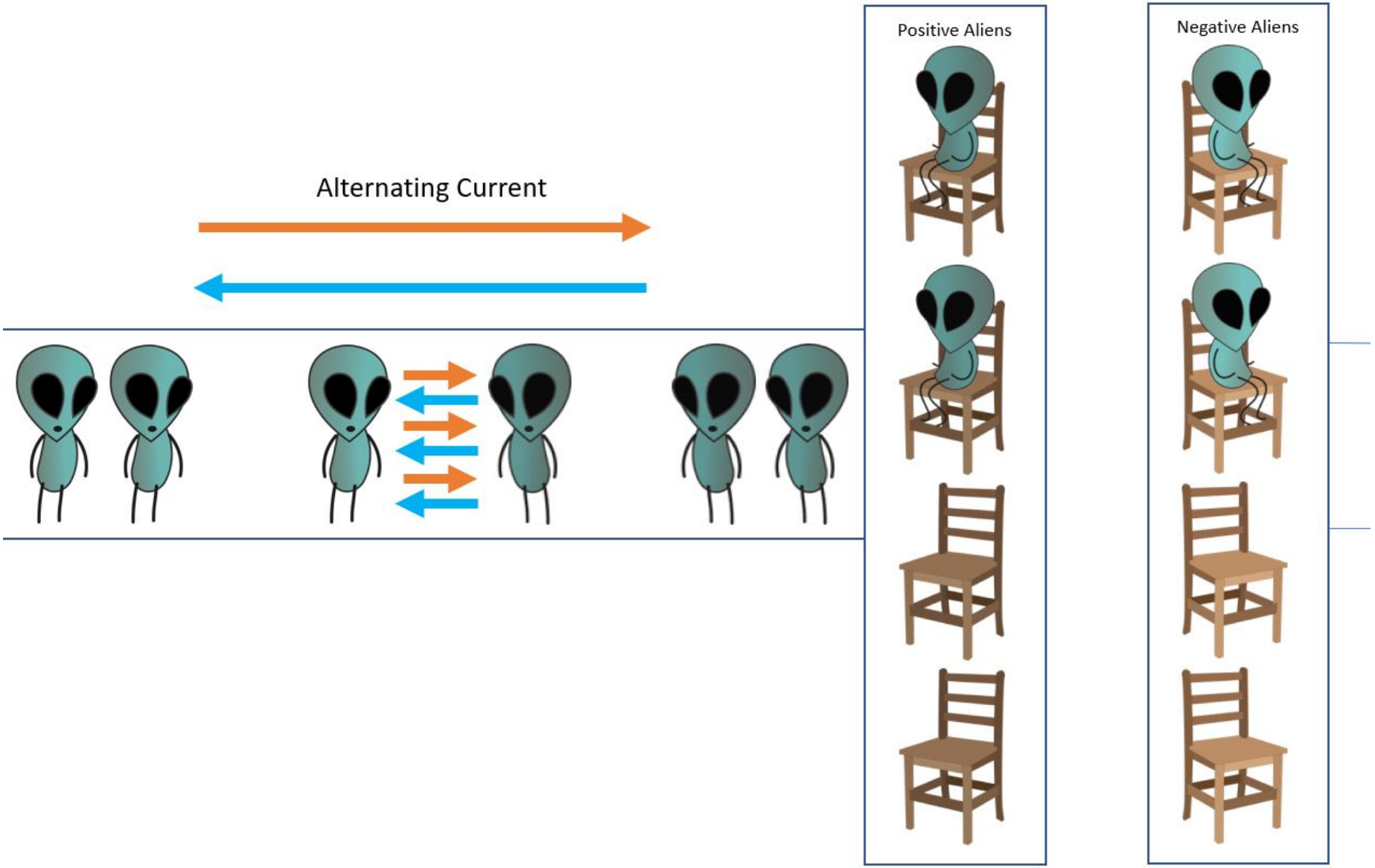
$$I = C\frac{\Delta V}{\Delta t}$$

And, therefore, the capacitor charges with a time constant (τ):

$$\tau = RC$$

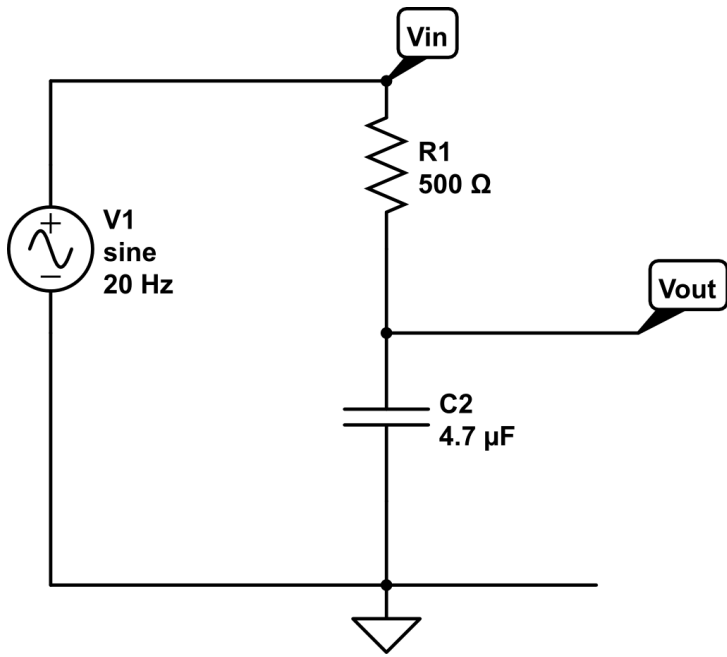


Alternating Current



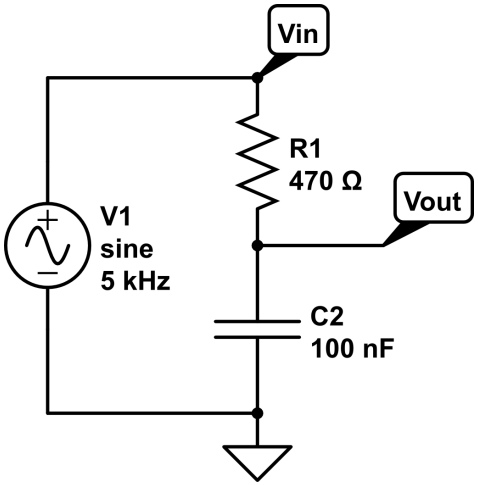
Low Pass Filter - cutoff frequency f_c

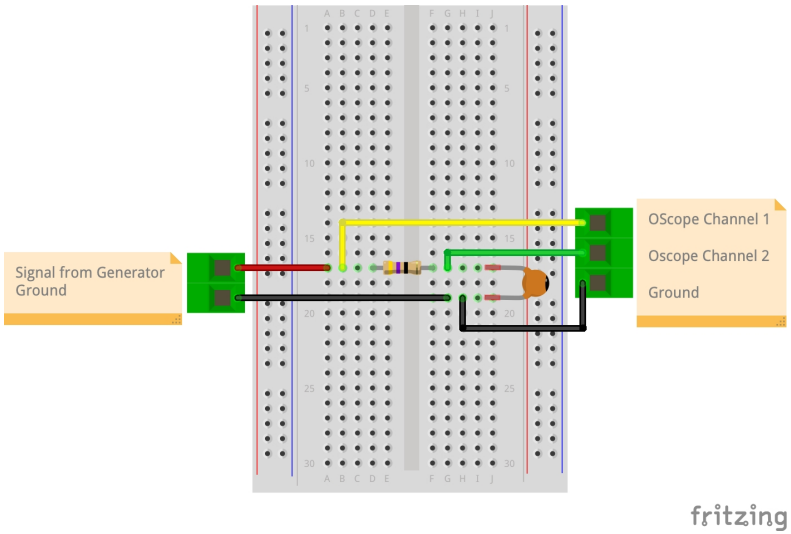
- At low frequencies, there is plenty of time for the capacitor to charge up to practically the same voltage as the input voltage.
- At high frequencies, the capacitor only has time to charge up a small amount before the input switches direction. The output goes up and down only a small fraction of the amount the input goes up and down. At double the frequency, there's only time for it to charge up half the amount.



$$f_c = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC}$$

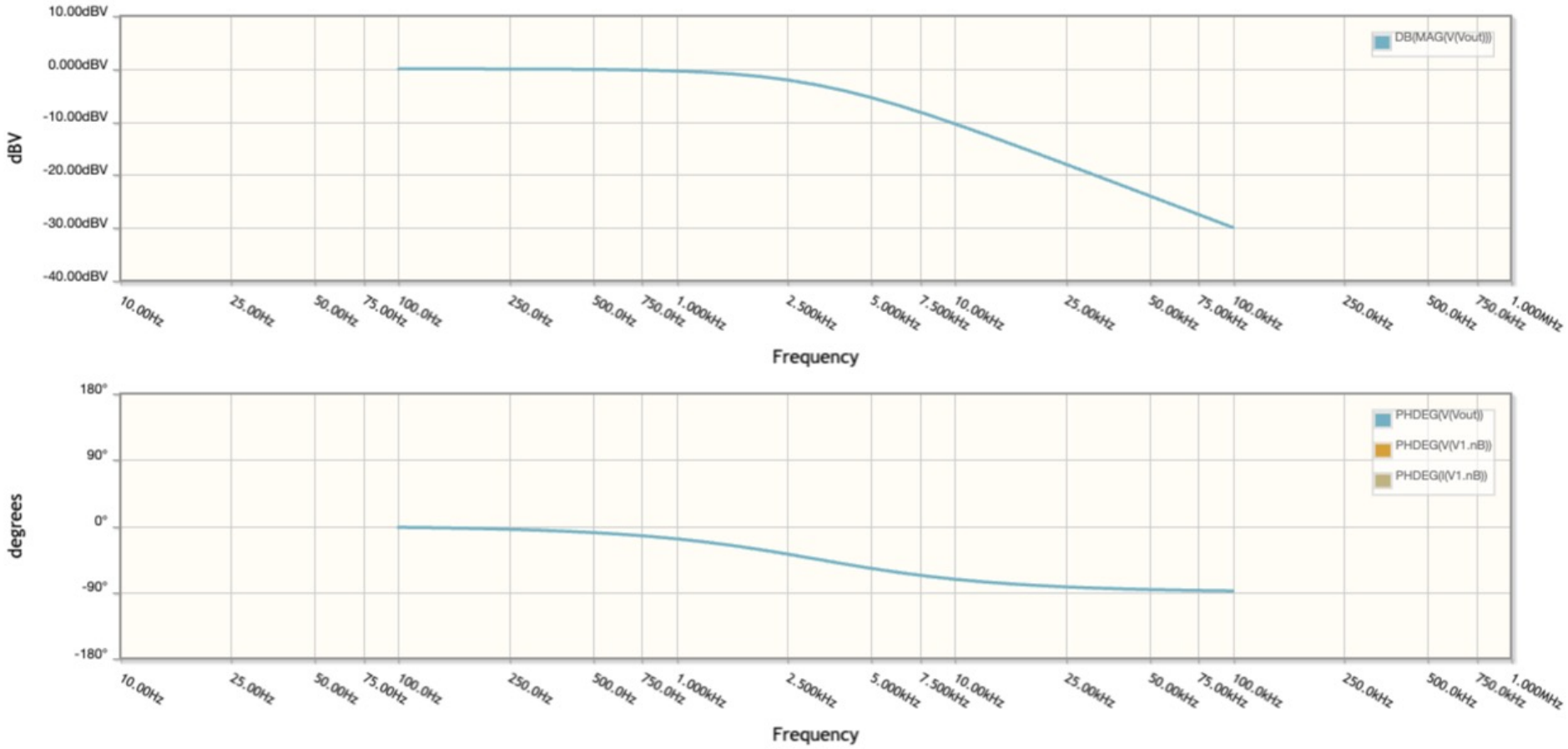
Assignment: Low Pass Filters





- Build a low pass filter as shown
- Set the waveform generator to:
  - Sine Wave
  - 5V
  - Sweep 100 Hz to 100MHz
- Record the input, output attenuation, and phase

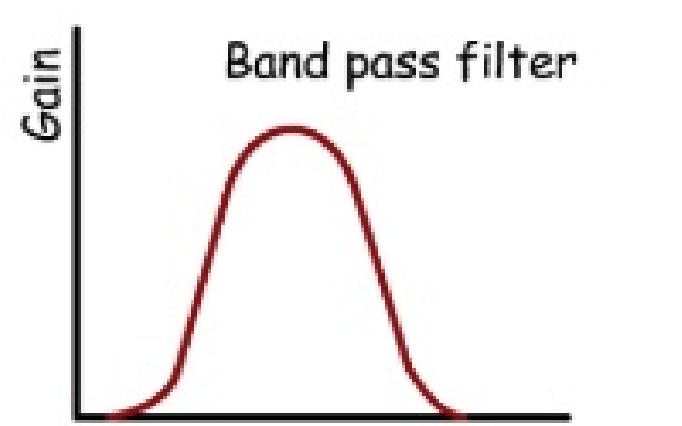
Low Pass Filter Response

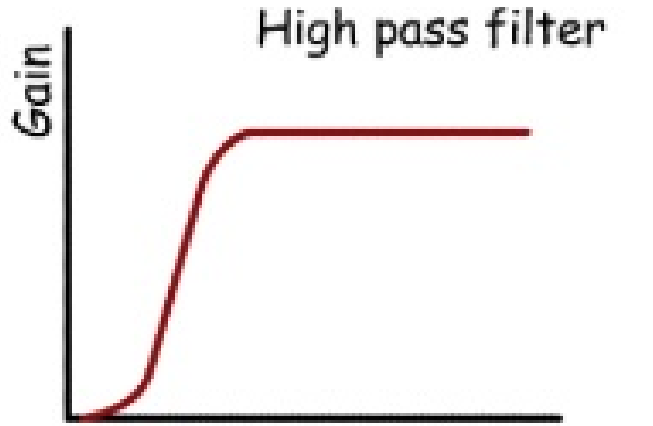


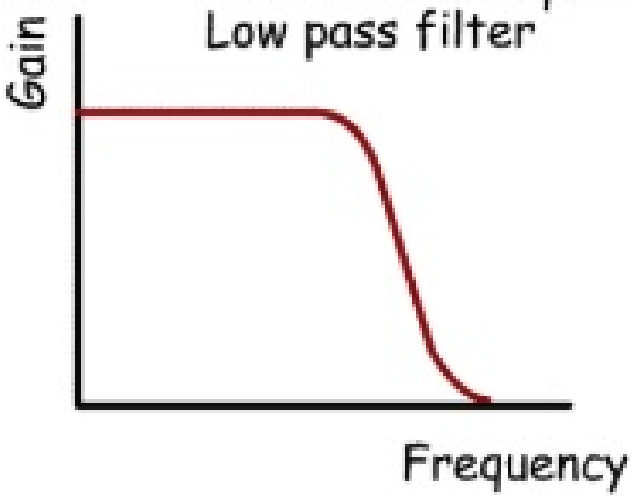
$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(500)(01.*10^{-6})} = 3.34kHz$$

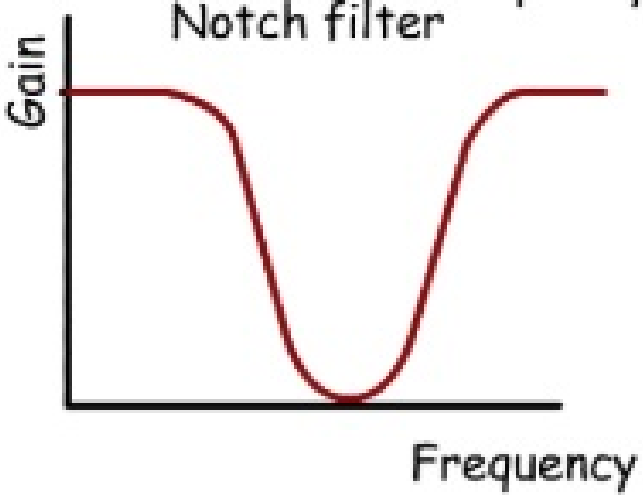


Other types of filters



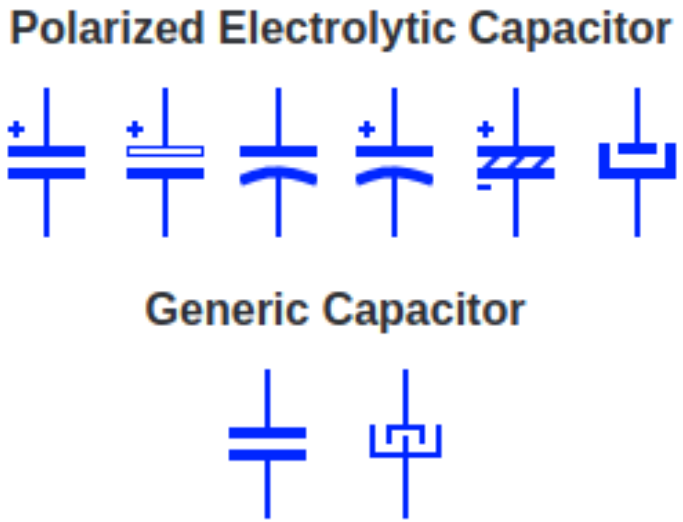




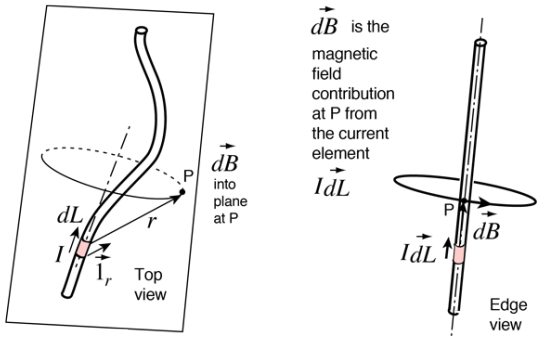


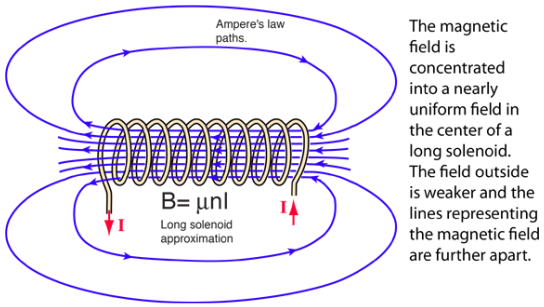
Capacitors - does it matter how they are placed

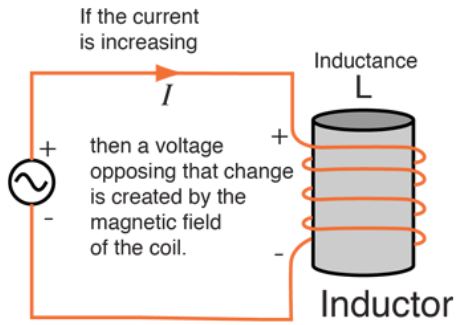
- Some types of capacitors (electrolytic and tantalum) are polarized (they have + and - terminals). This is due to how the dielectric film has been deposited. The reverse polarity leads to degradation of the dielectric.
- Other capacitors (ceramic and film) do not have a polarity and can be installed in either direction.



Inductors

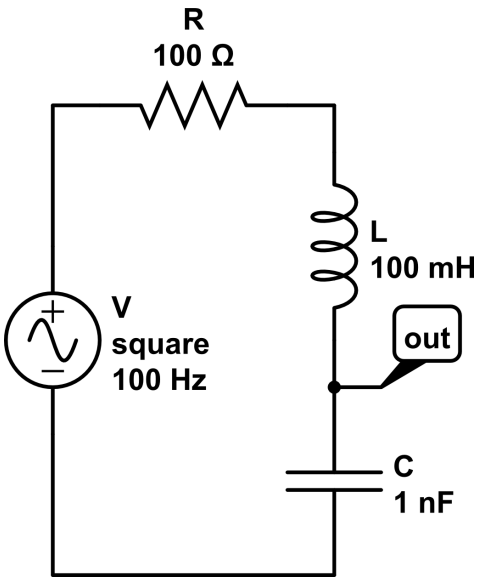


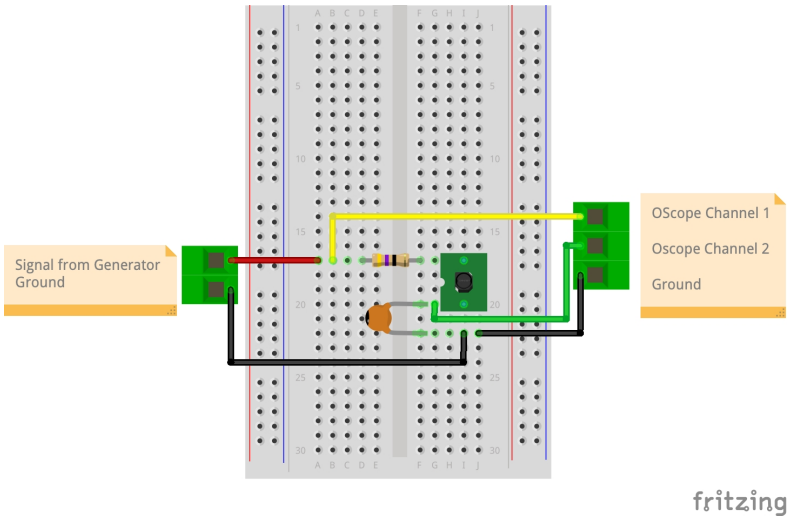




- Current flowing in a wire produces a magnetic field (B) around the wire (from Ampere's Law).
- Wire wrapped into a coil produces a magnetic field that resembles a bar magnet through the center of the coil.
- Also, in a coil, this magnetic field produces an effect known as Inductance (L) that opposes changes in electric current.

Assignment: RLC Oscillator

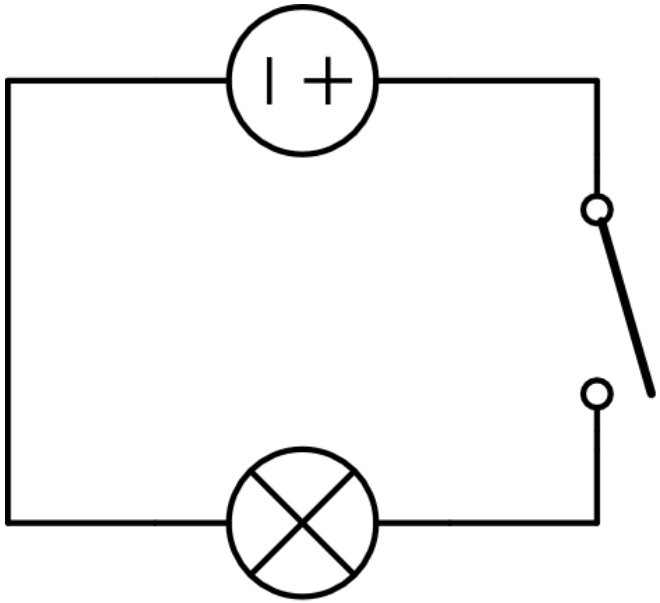




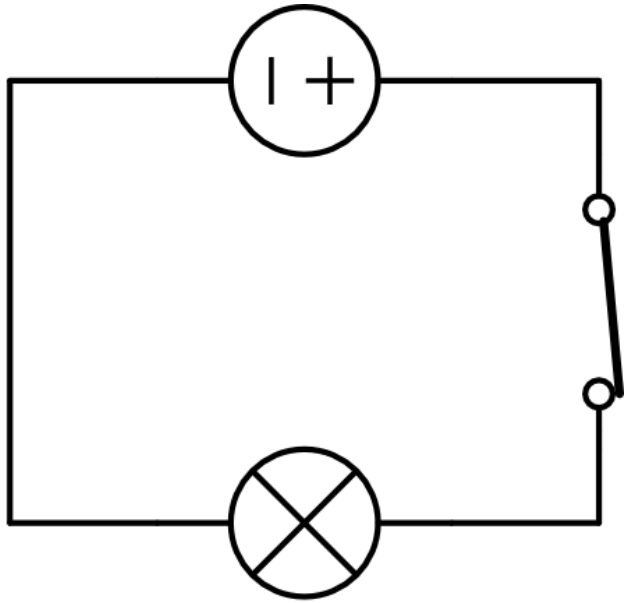
- Build the oscillator as shown
- Set the waveform generator to:
  - Square Wave
  - 5V
  - Frequency = 100 Hz
  - Duty Cycle to 0.1
- Calculate the period of the oscillations.
- Calculate the dampening over time.

Switches and Relays

Switches

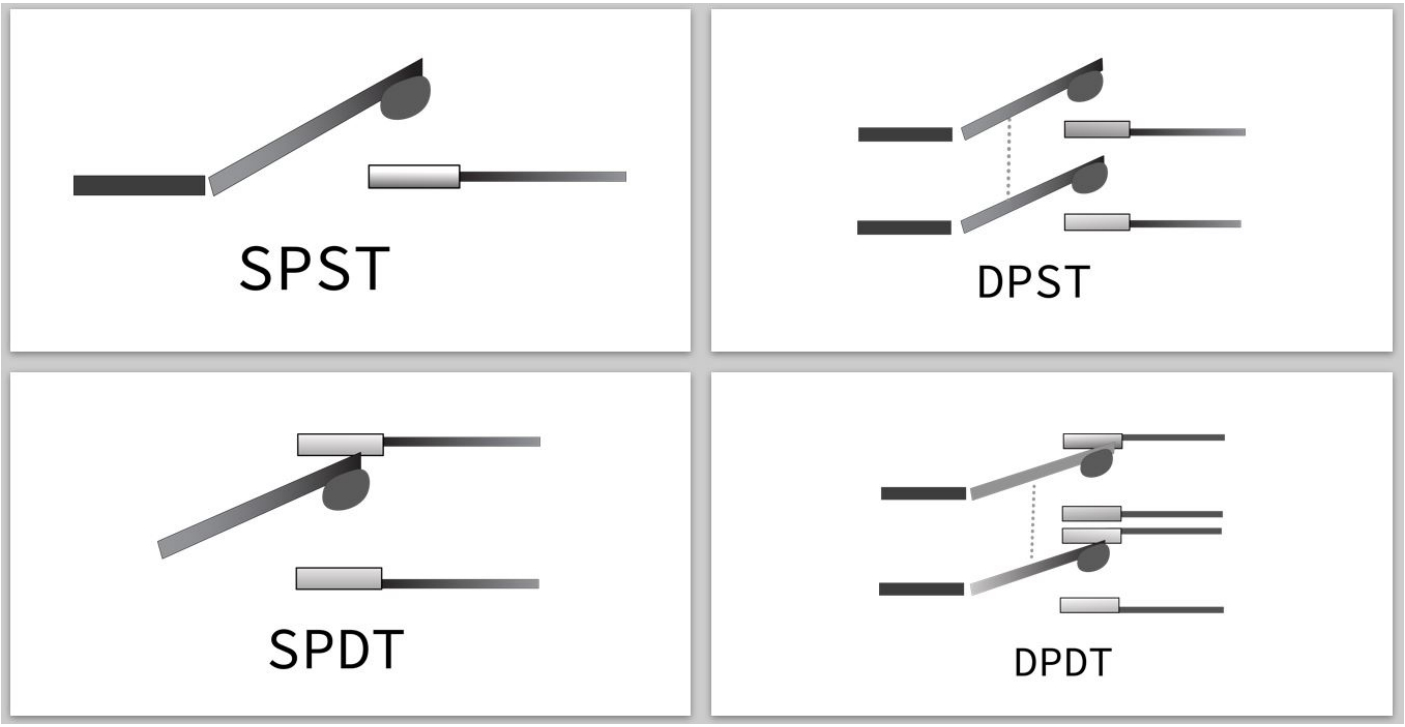


Lamp Off



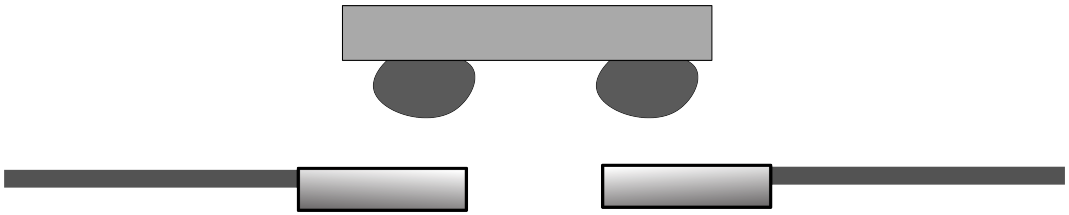
Lamp On

Poles and Throws



- Poles indicates the number of circuits that one switch can control for one operation of the switch.
- Throws indicates the number of contact points.

A Button - SPST



SPST

double break



Maxwell's Equations

- Gauss's Law

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

- Gauss's Law for Magnetism

$$\nabla \cdot B = 0$$

- Faraday's Law

$$\nabla \times E = -\frac{\delta B}{\delta t}$$

- Ampere-Maxwell Law

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\delta E}{\delta t}$$



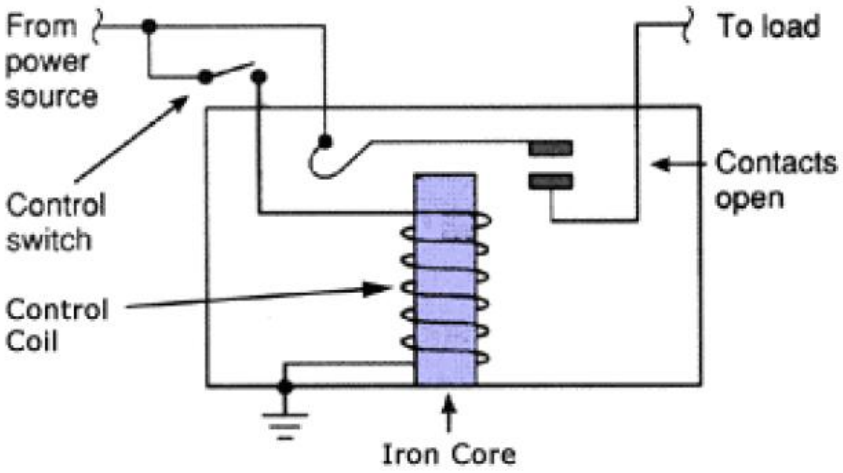
Maxwell's Equations - What They Mean

- Gauss's Law: Electric charges are "sources" or "sinks" for electric fields
- Gauss's Law for Magnetism: There are no magnetic monopoles
- Faraday's Law: A changing magnetic field induces current
- Ampere-Maxwell Law: Moving charges and changing electric fields both create circulating magnetic fields.

Permittivity and Permeability:

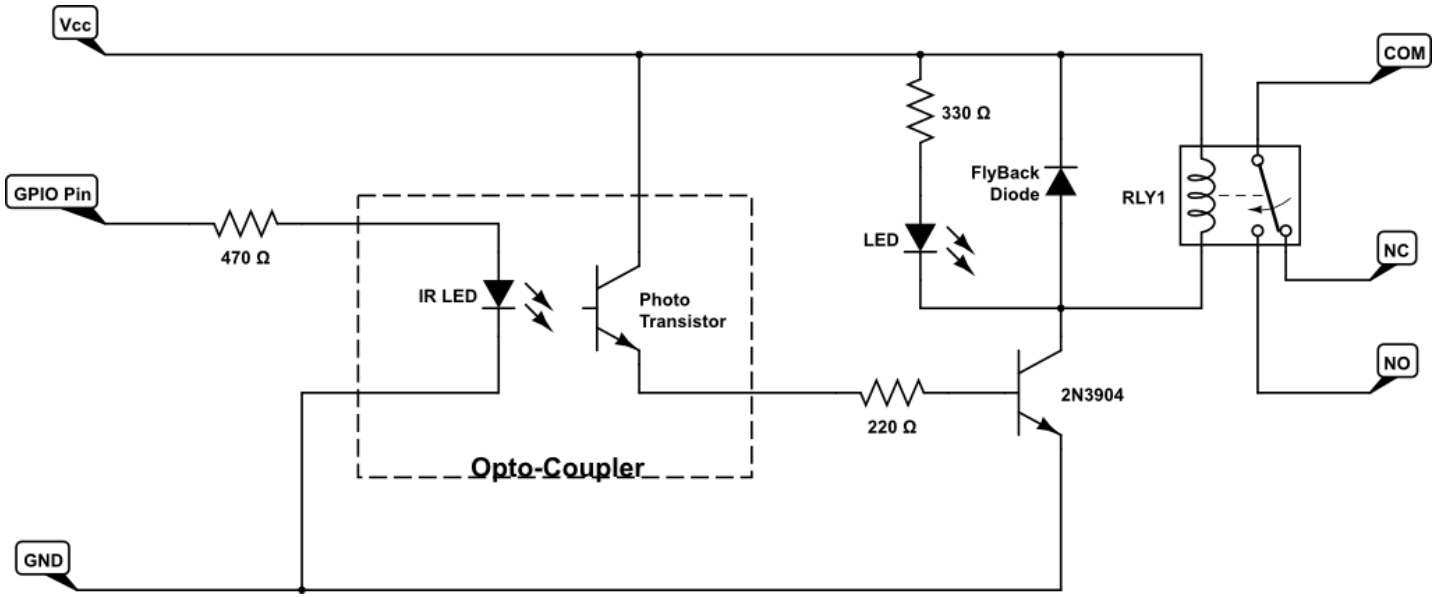
- Permittivity of free space: $\epsilon_0 = 8.85 \times 10^{-12}$ (F/m)
- Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}$ (H/m)

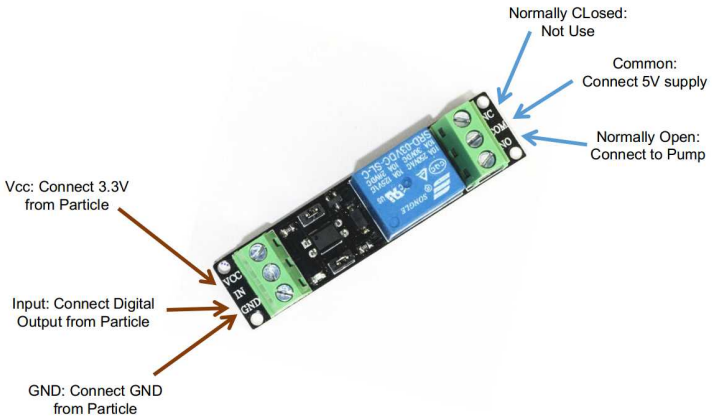
Relays



- When the control terminal has current, the coil around the iron core creates a magnetic field
- The magnetized core attracts a cantilevered arm closing the circuit to the load
- When current is stopped, the core is no longer magnetized and a spring opens the contact.

Optocoupled Relay





- Optocoupler isolates the relay load (which could be up to 240V) from the controller electronics.

The 35144 Relay in the Vacuum Systems

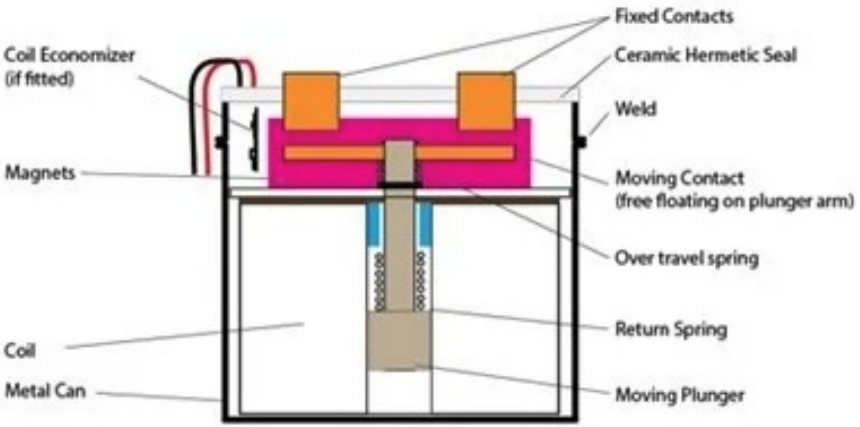
Double Pole, Double Throw Relay





Contactor

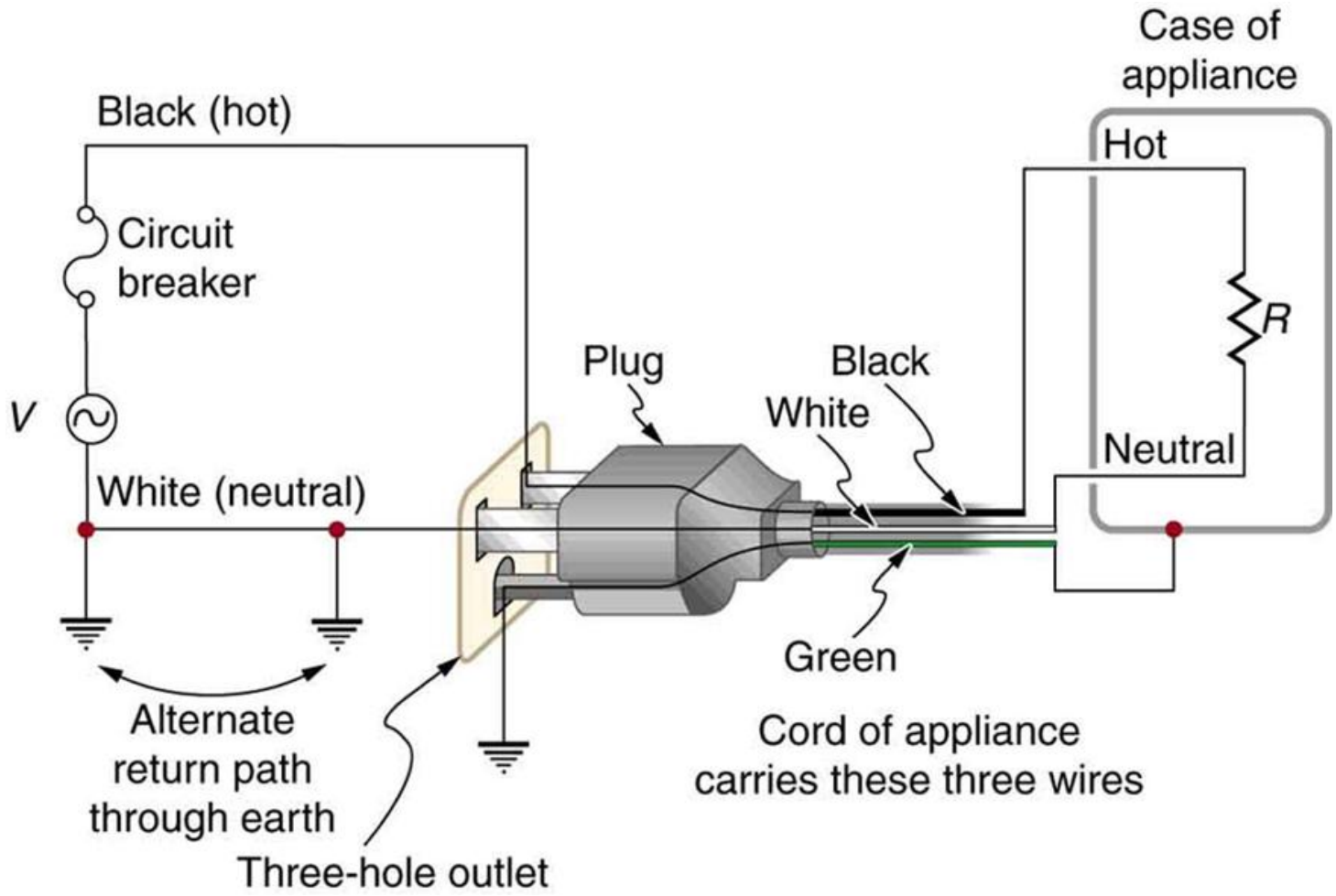




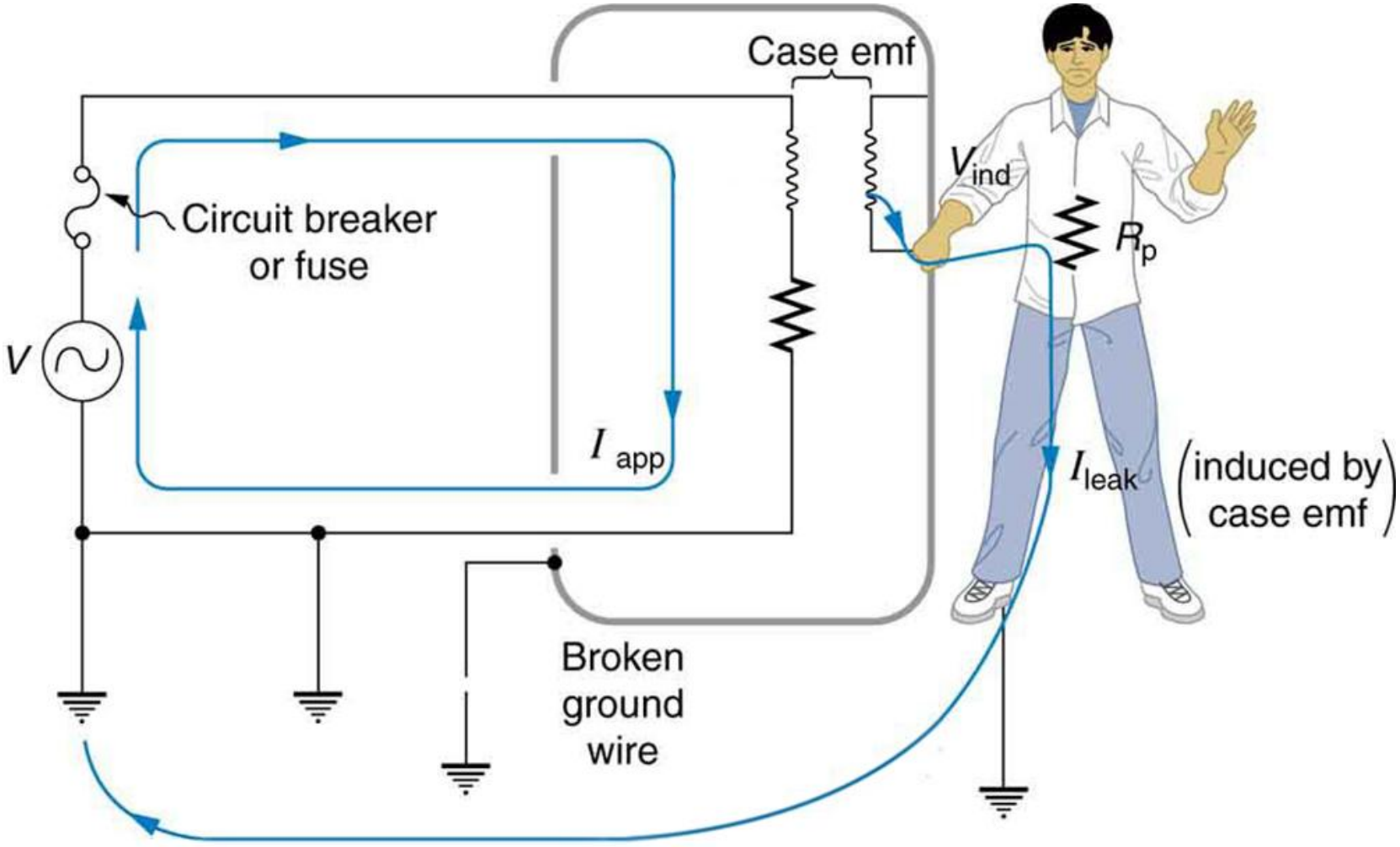
Difference between contactor and other relays

- Higher current, voltage, and power ratings
- Often join 2 poles without a common circuit between them
- Built in arc suppression, allowing them to handle high currents, such as motor starting inrush currents
  - Gas filled Contactors
  - Magnetic Blowout Contactors

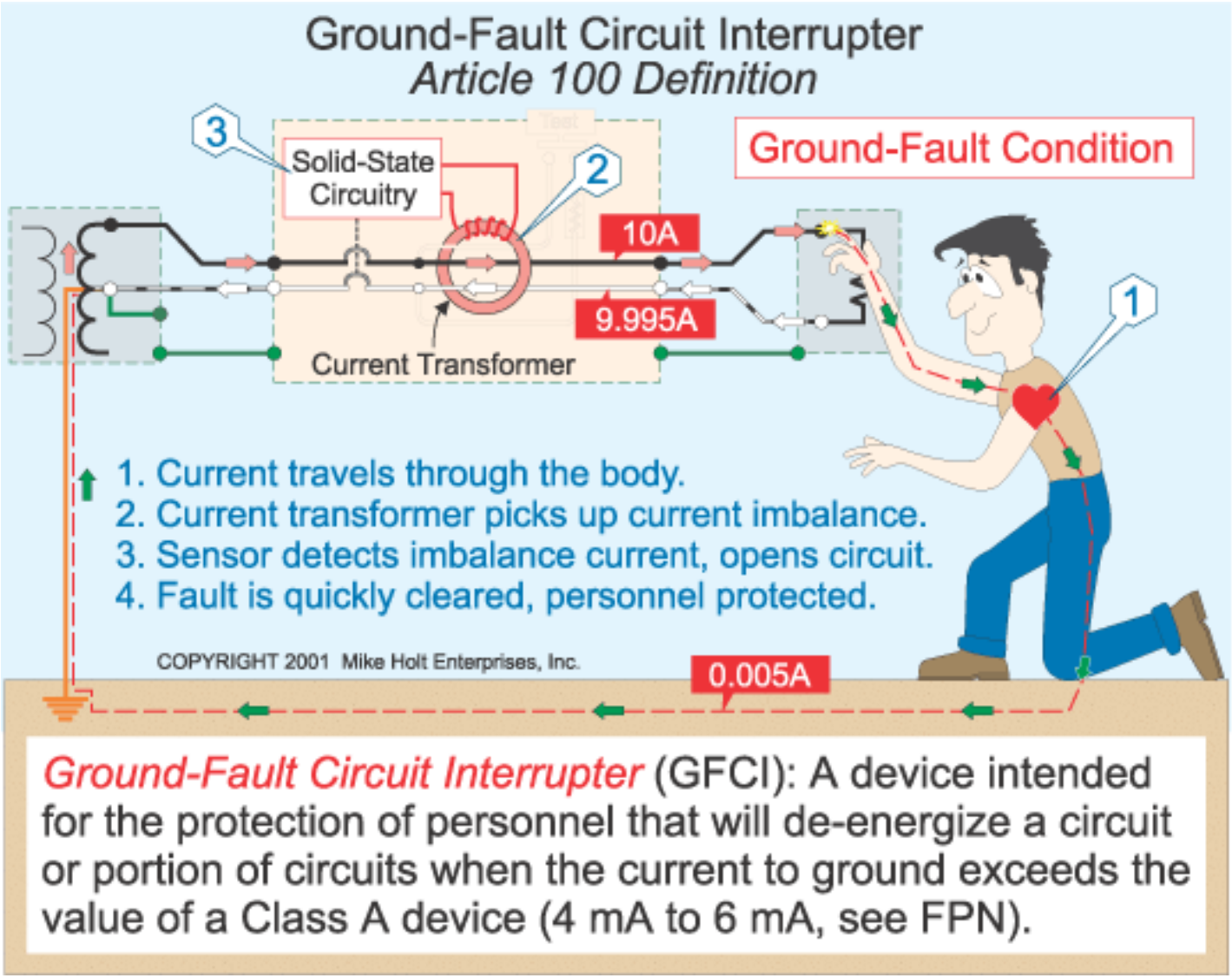
Electrical Safety - Three Wires



Electrical Safety - Electric Shock

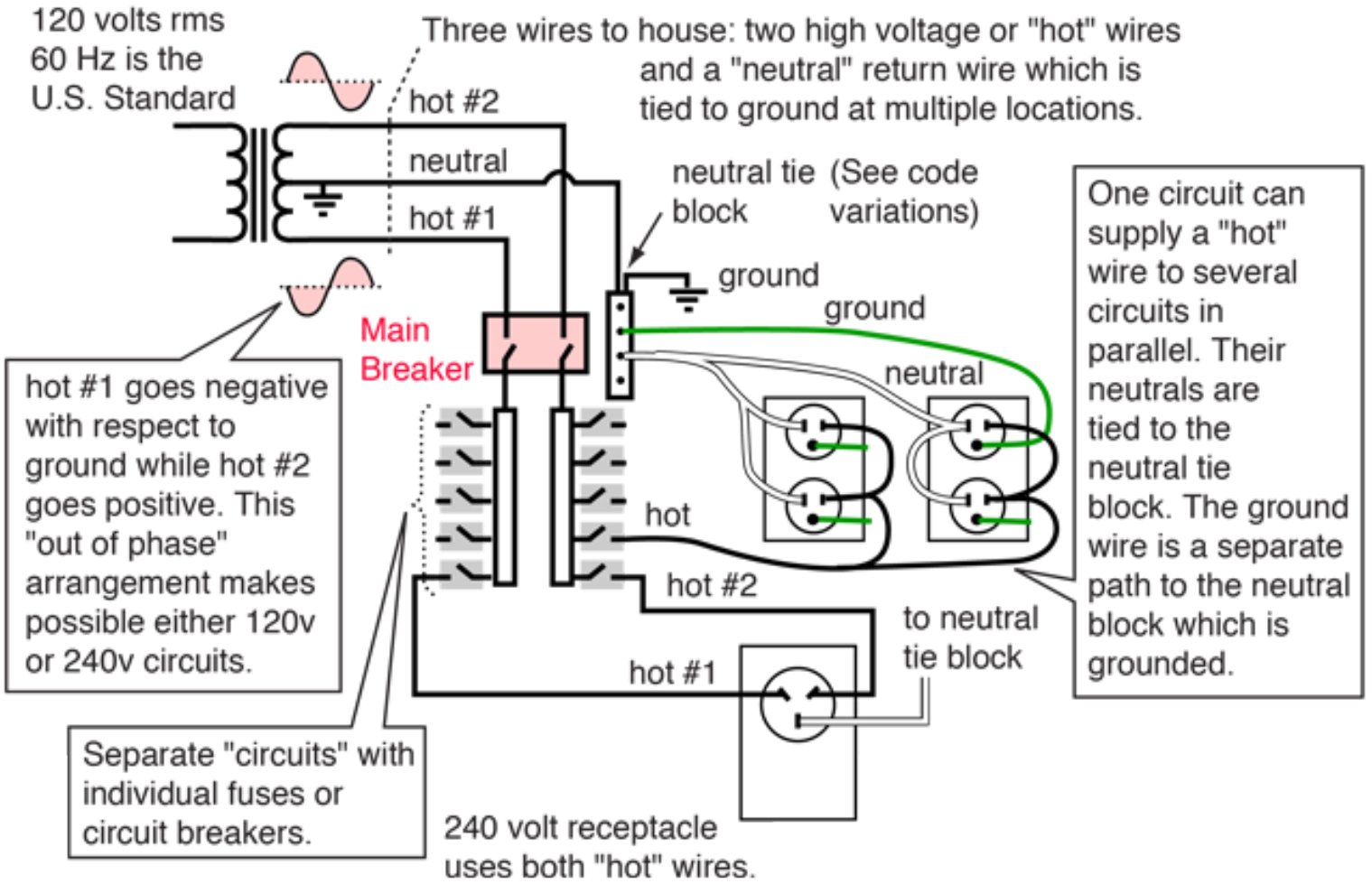


Electrical Safety - GFCI



Ground-Fault Circuit Interrupter (GFCI): A device intended for the protection of personnel that will de-energize a circuit or portion of circuits when the current to ground exceeds the value of a Class A device (4 mA to 6 mA, see FPN).

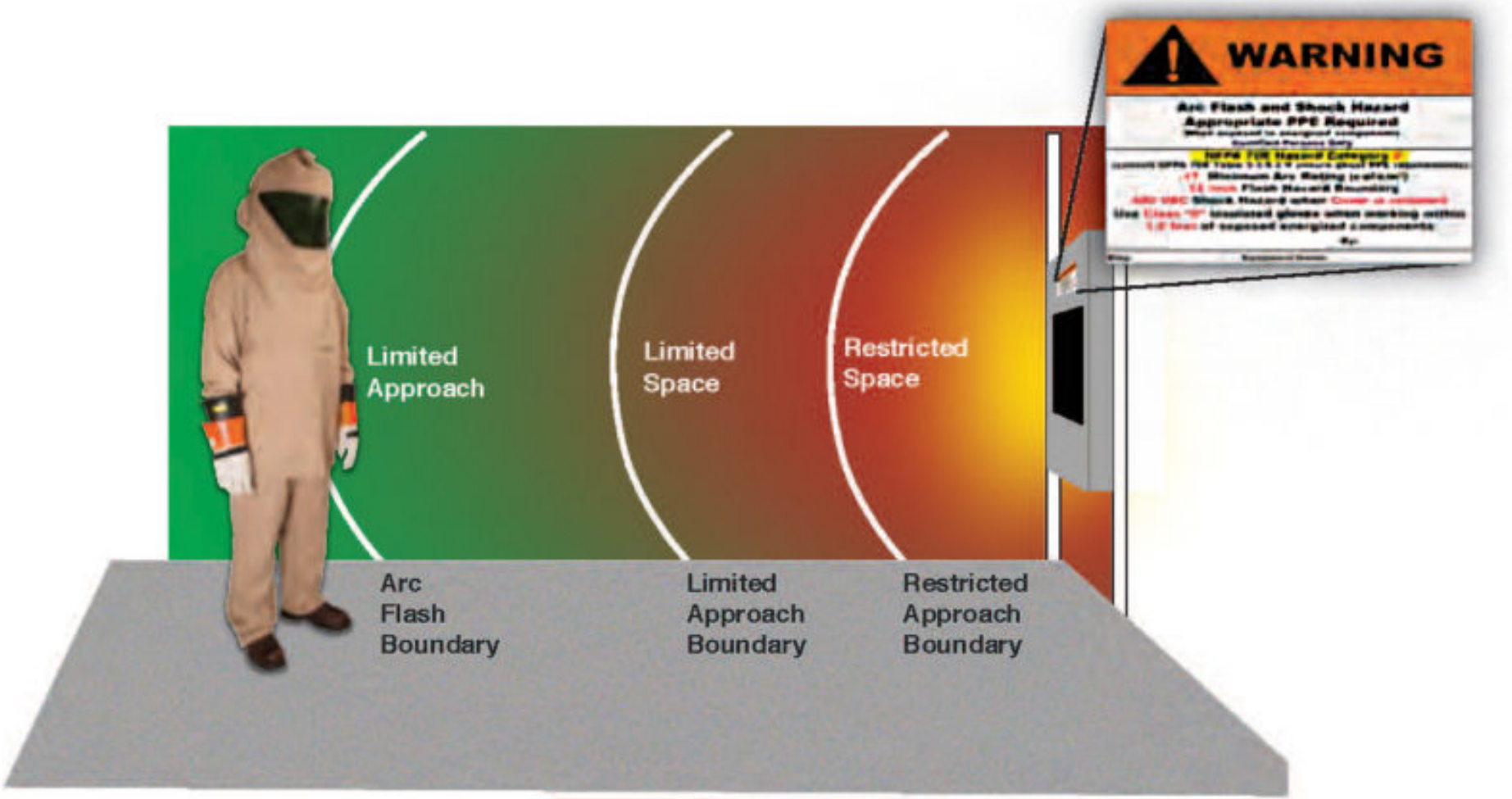
Electrical Safety - 240V



Arc Flash



Arc Flash


WARNING
Limited Approach
Limited Space
Restricted Space
Arc Flash Boundary
Limited Approach Boundary
Restricted Approach Boundary

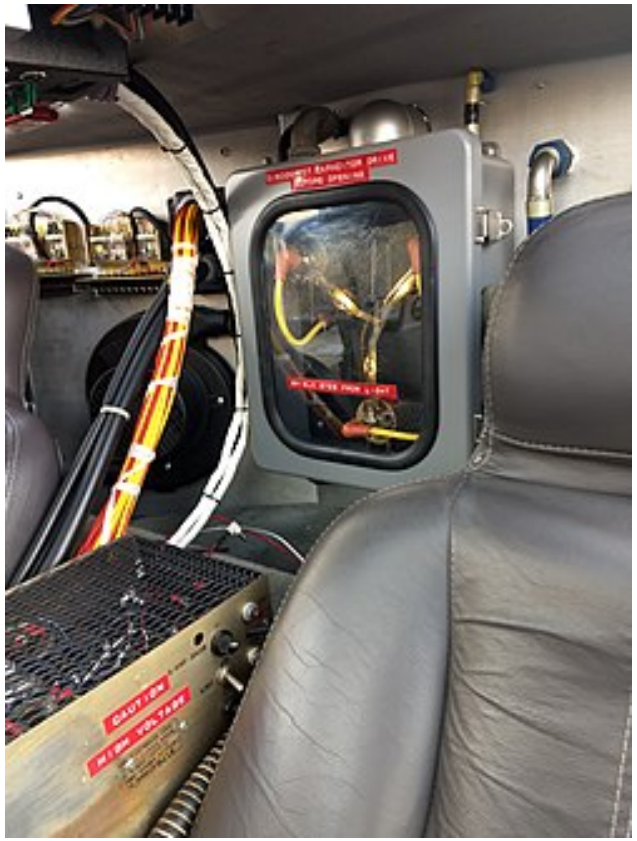
Three Point Check



1. Verify the test tool works properly when the function switch is placed to "voltage" by testing for voltage on a known energized source, or by using an electronic proving unit, and observing the correct reading on the meter face.
2. Test the circuit to be verified by measuring phase-to-phase and phase-to-ground across all phases. Zero energy must be indicated.
3. Ensure the test tool still indicates voltage properly by placing the test probes, once again, on a known, energized source or the electronic proving unit.

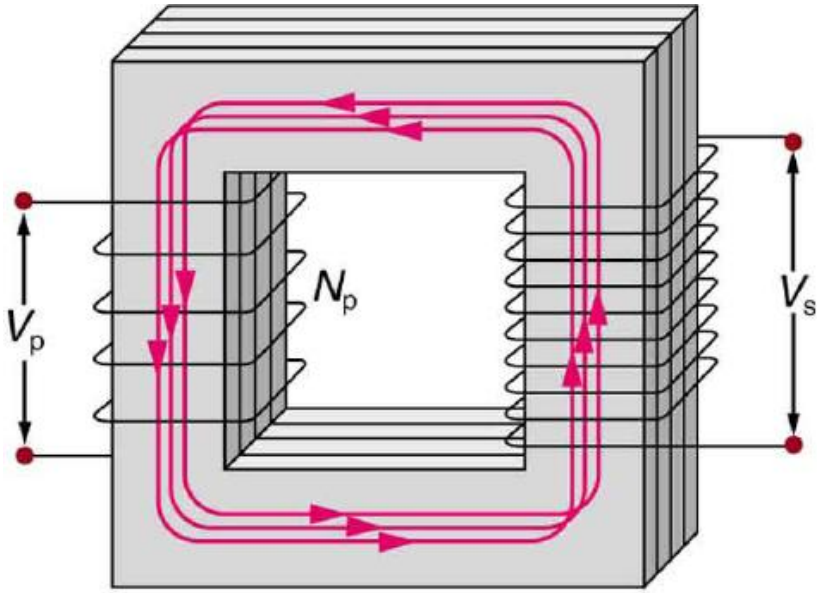
Transformers and Rectifiers

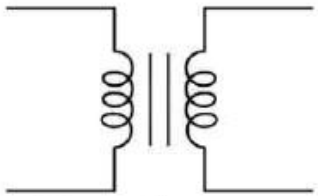
Magnetic Flux



Magnetic Flux (Φ) is the amount of magnetic field lines per unit area.

Transformer





Transformer symbol

From Faraday's Law:

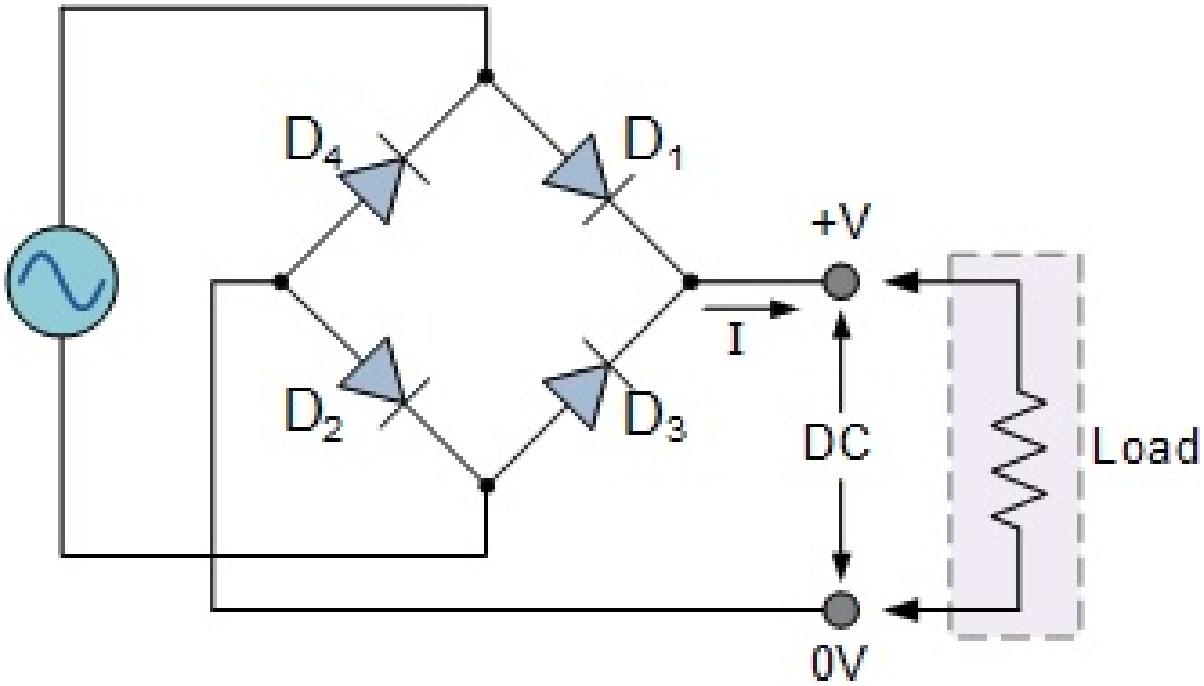
$$V = -N\frac{\Delta\Phi}{\Delta t}$$

Because the change in flux is the same on both sides of the transformer:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

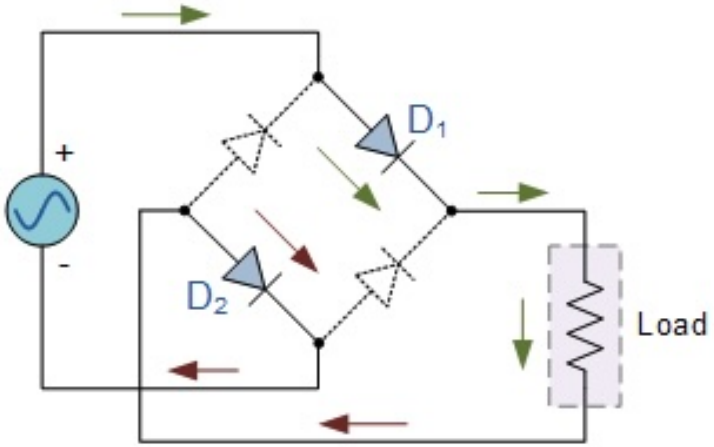
The voltage change is a result of the different number of windings

Full Wave Rectifier

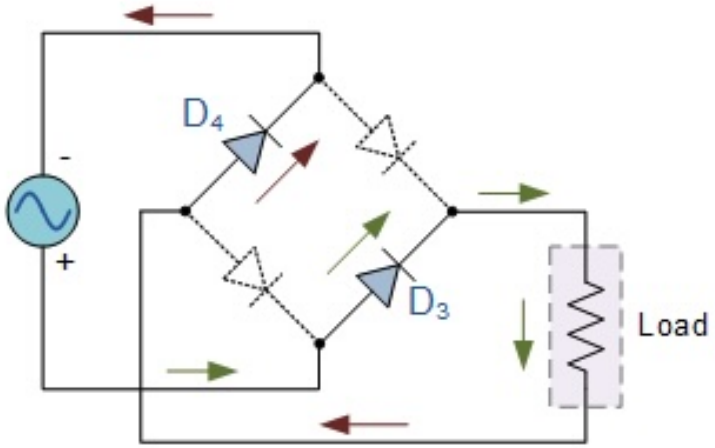


Full Wave Rectifier in Action

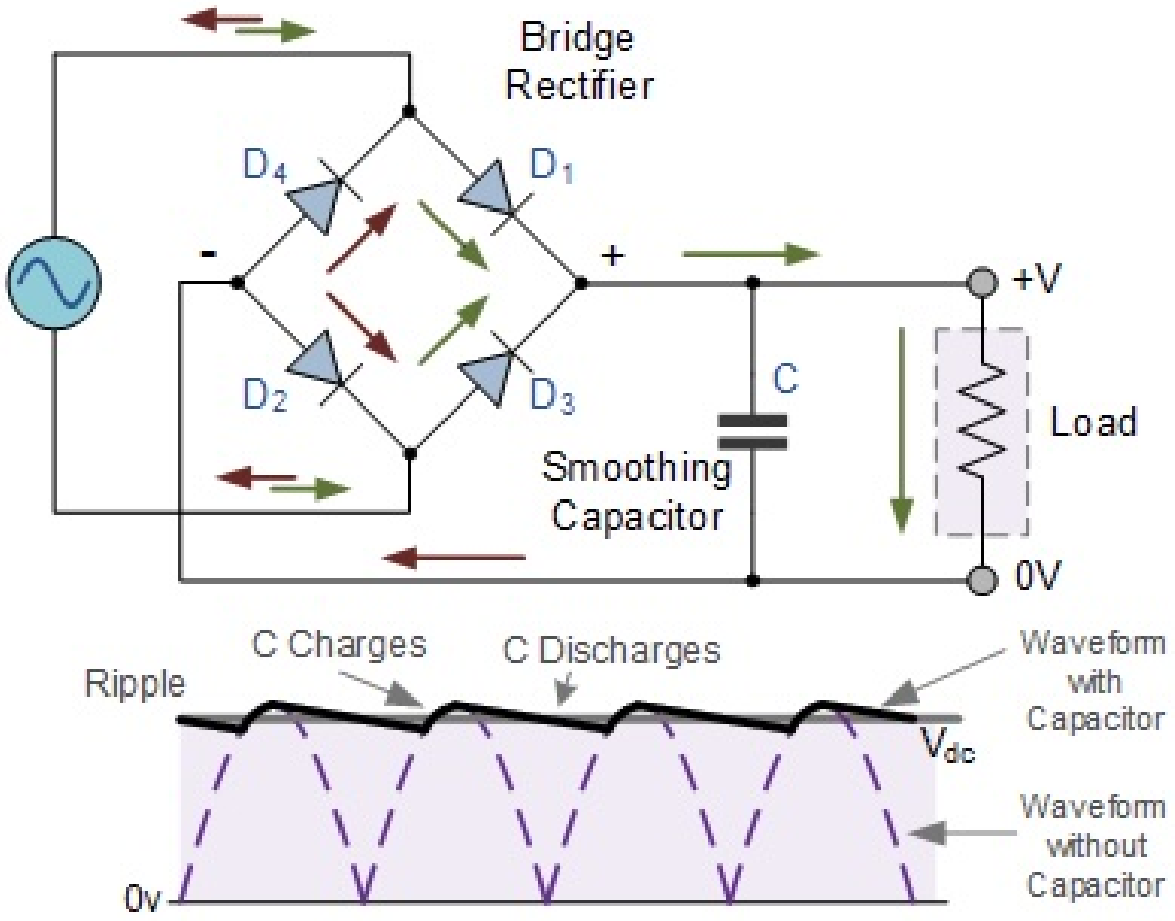
Positive Half Cycle



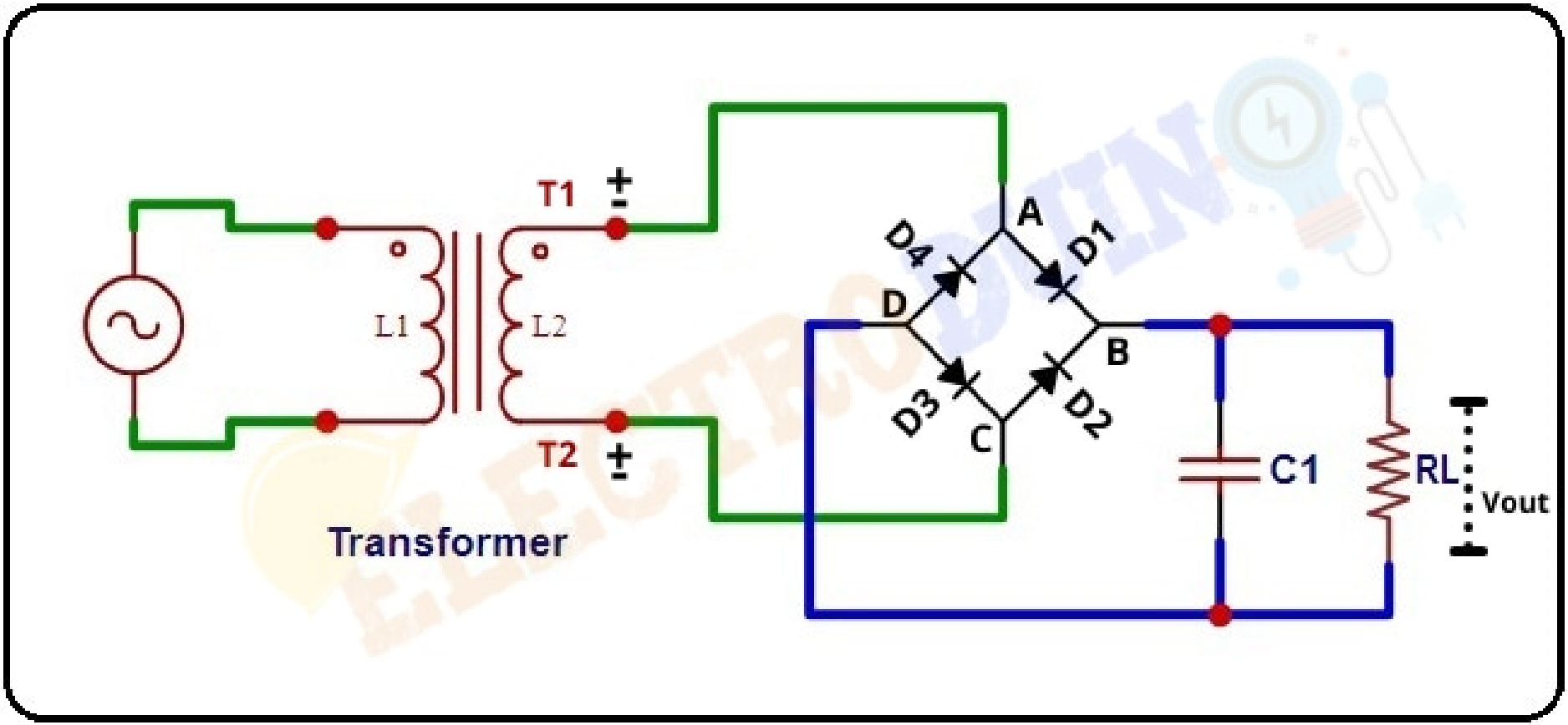
Negative Half Cycle



Full Wave Rectifier with Smoothing Capacitor



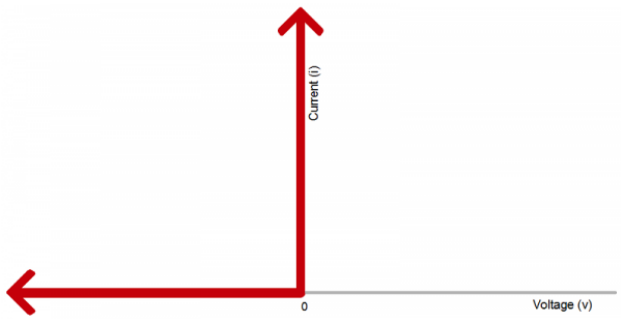
Putting it All Together

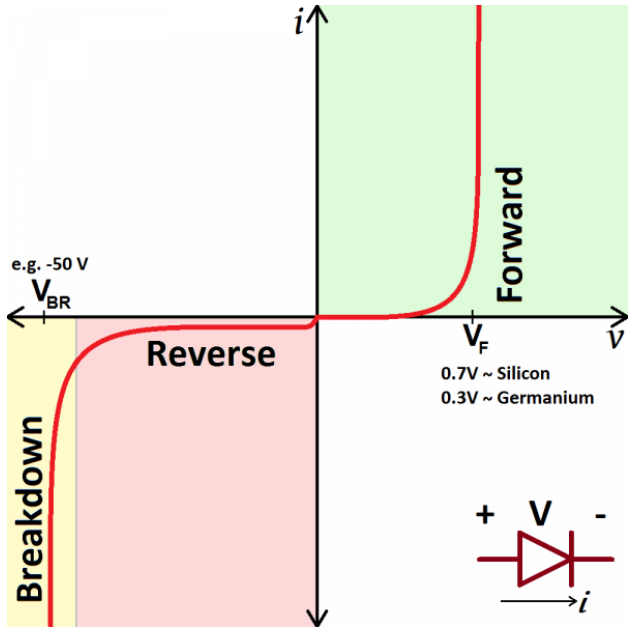


Diodes and Photodetectors

Diodes

The key function of a diode is to control the direction of current-flow. Current passing through a diode can only go in one direction, called the forward direction. Current trying to flow the reverse direction is blocked.

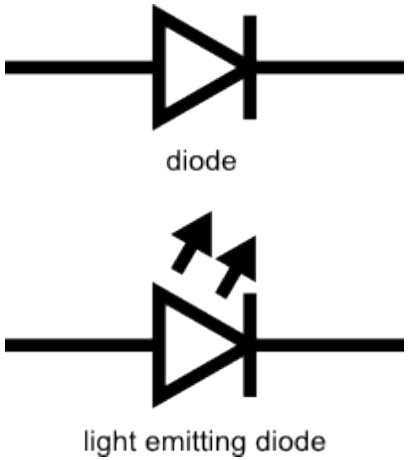




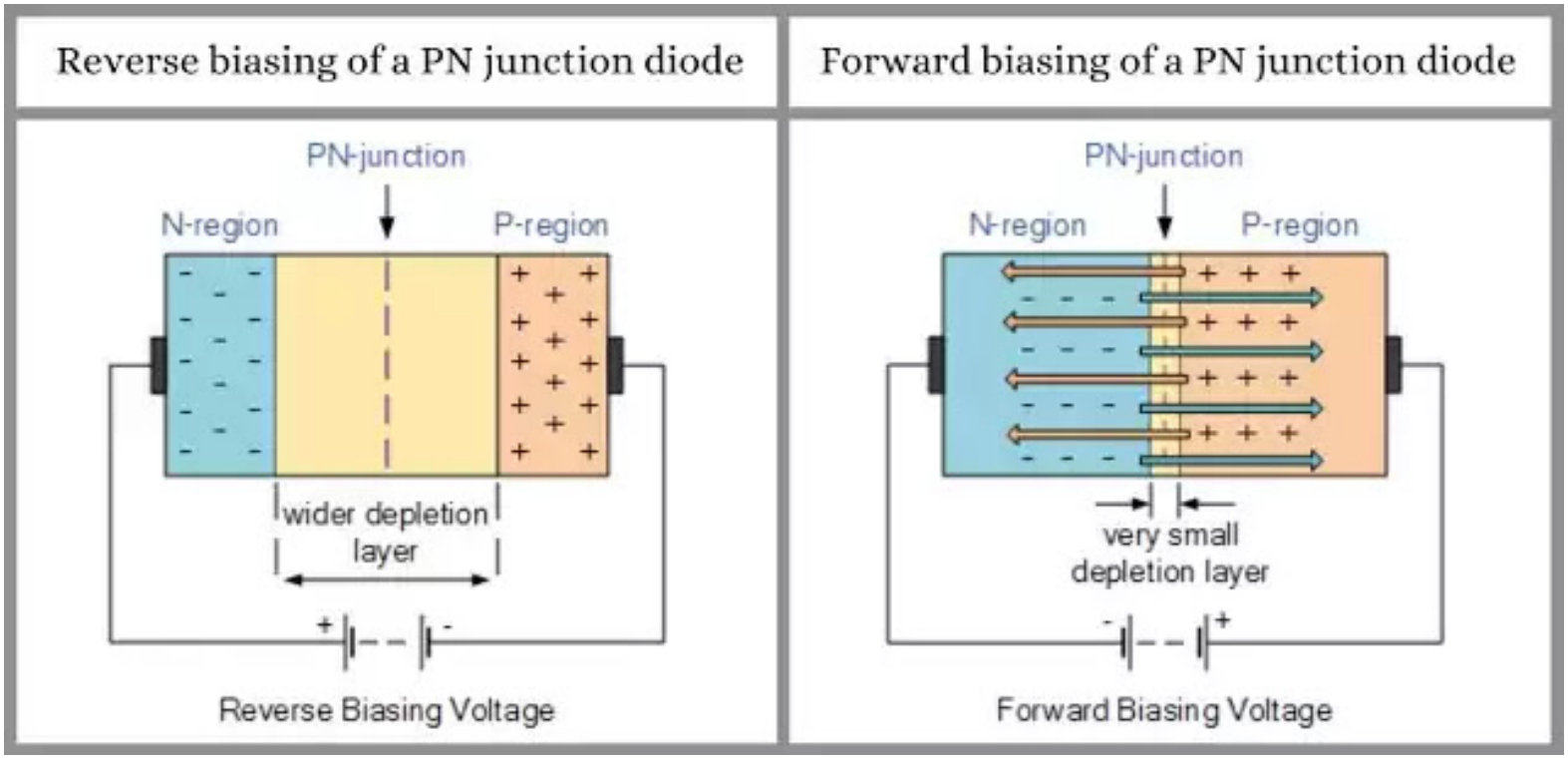
A non-ideal diode needs a small forward voltage (V_F) to turn on and will fail (breakdown) with a large negative voltage (V_{BR}).

Light Emitting Diodes

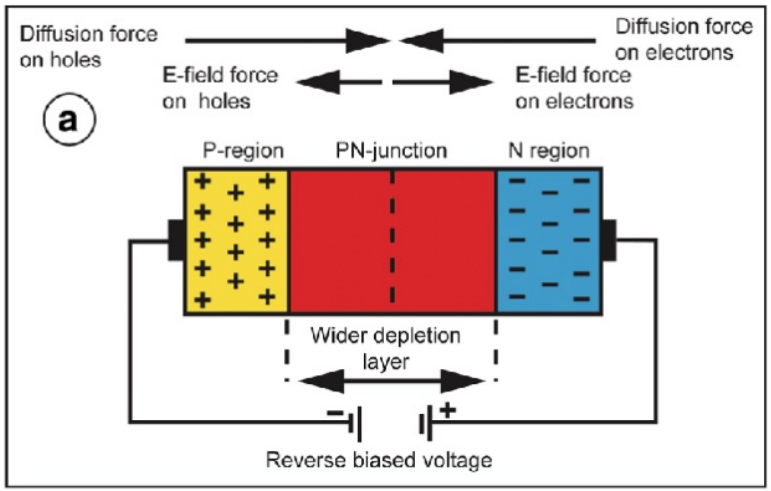
LEDs (that's "ell-ee-dees") are a particular type of diode that convert electrical energy into light.

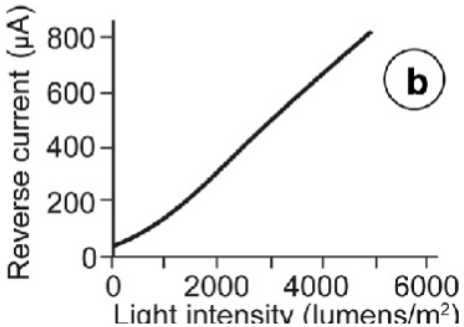


Diodes in Reverse and Forward Bias

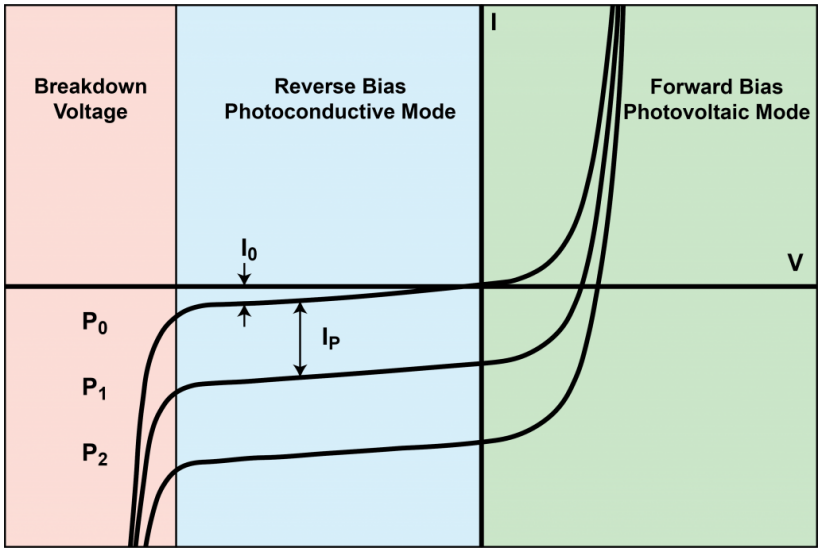


Photodiode

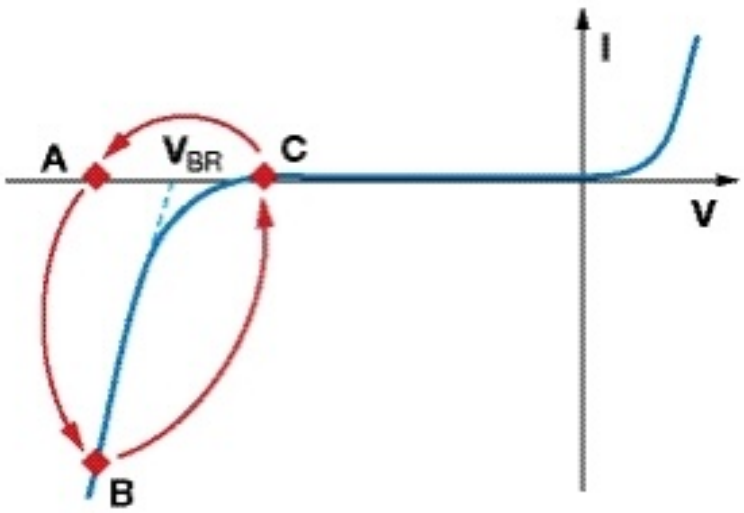

Diffusion force on holes
Diffusion force on electrons
E-field force on holes
E-field force on electrons
a
P-region
PN-junction
N region
Wider depletion layer
Reverse biased voltage


Reverse current (μA)
800
600
400
200
0
b
0
2000
4000
6000
Light intensity (lumens/m²)


c


Breakdown Voltage
Reverse Bias Photoconductive Mode
Forward Bias Photovoltaic Mode
I
V
I0
P0
P1
P2
IP

Avalanche Photodiode (APD)



- Operates in reverse direction and biased slightly beyond the breakdown threshold voltage V_{BR} (Geiger Mode)
- An active quenching circuit limits the current through the APD in order to avoid destruction and lowers the bias voltage below V_{BR}
- There is a deadtime after quenching
- In addition to photo-generated events, Dark Counts can happen spontaneously

Operational Amplifiers

Keysight Oscilloscope



Keysight Waveform Generator



Keysight 3-Channel Power Supply



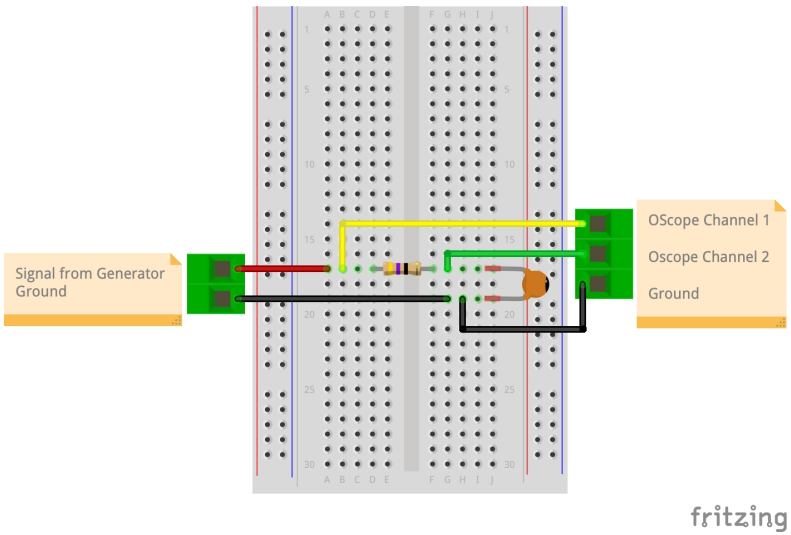
Assignment: Oscilloscope Basics





- Connect Waverform Generator Output 1 to Oscilloscope Input 1
- On the Waveform Generator:
  - Select waveform → Sine
  - Set amplitude to 5V
  - Set frequency to 10 kHz
  - Turn on Channel 1
- On the Oscilloscope:
  - Select Channel 1
  - Adjust vertical and horizontal until you see a standing sine wave
- Experiment with adjusting amplitude, frequency, and waveform type
- Note how the data in upper left, upper right, and bottom of oscope changes

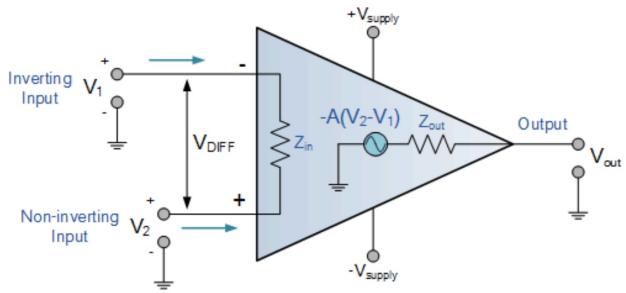
Assignment: Low Pass Filter

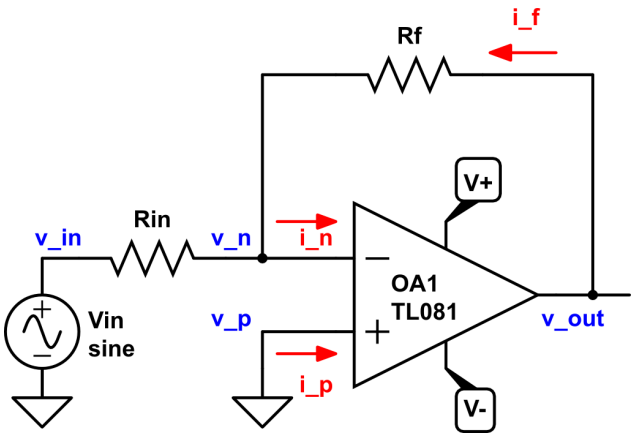


Select sine and sweep on the waveform generator

- Amplitude 500mV
- Sweep On
- Log
- Frequency Start 100Hz
- Frequency Stop 100kHz
- Sweep Time 30 seconds
- With oscilloscope estimate change in gain and phase

Ideal Op Amp





- Op Amp amplifies the difference in voltage between the two inputs
- Ideal OpAmp Assumptions
  - Amplification is infinite ($> 20k$)

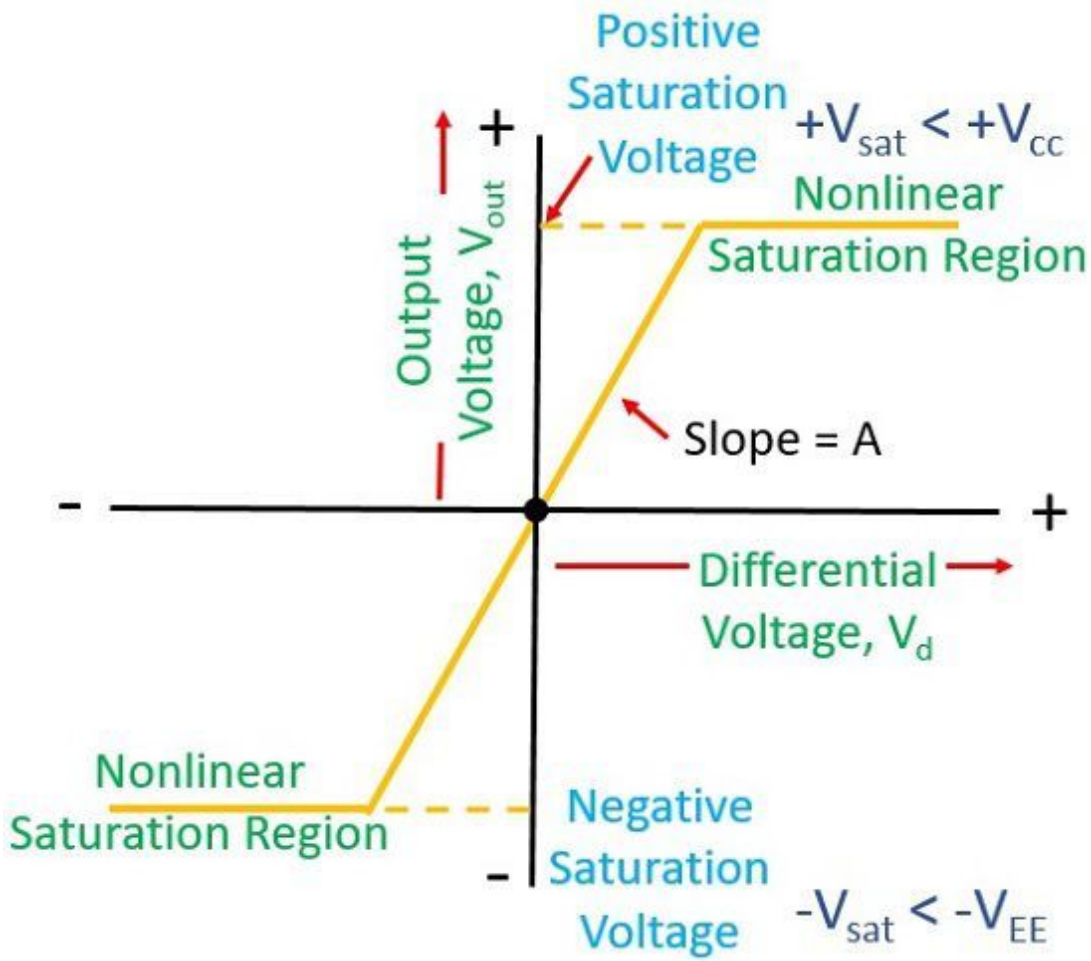
$$v_n = v_p$$

  - Input impedance is infinite

$$i_n = i_p = 0$$

  - Output impedance is zero

OpAmp in saturation



Op Amp Circuits

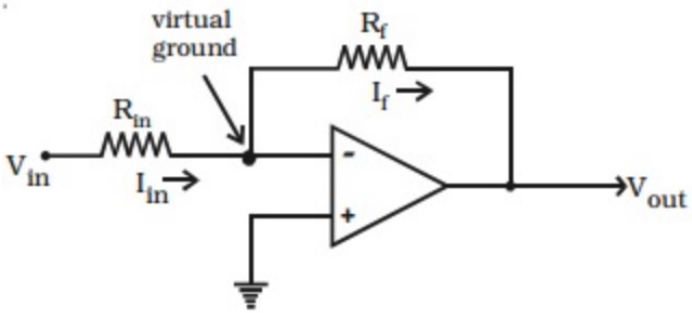


Fig Inverting amplifier

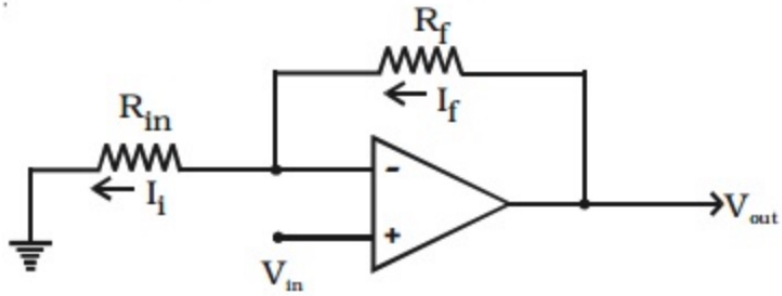


Fig Non-inverting amplifier

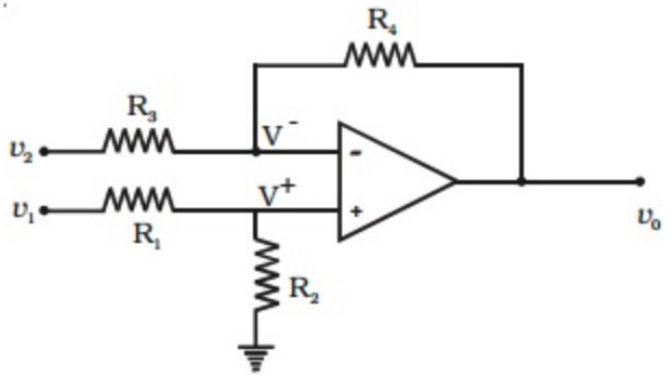


Fig Difference amplifier

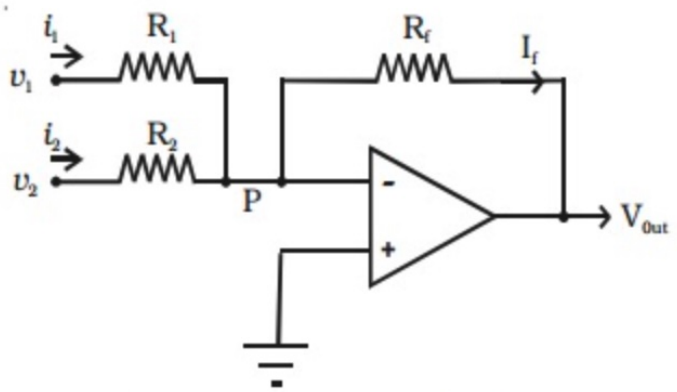
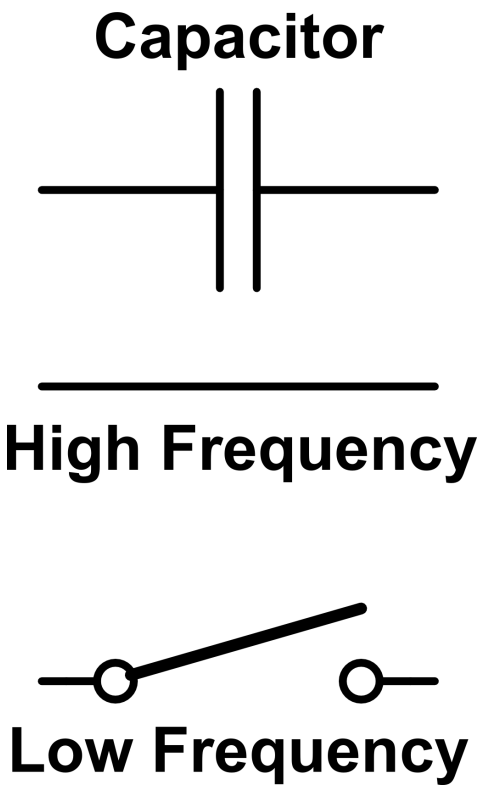


Fig Summing amplifier

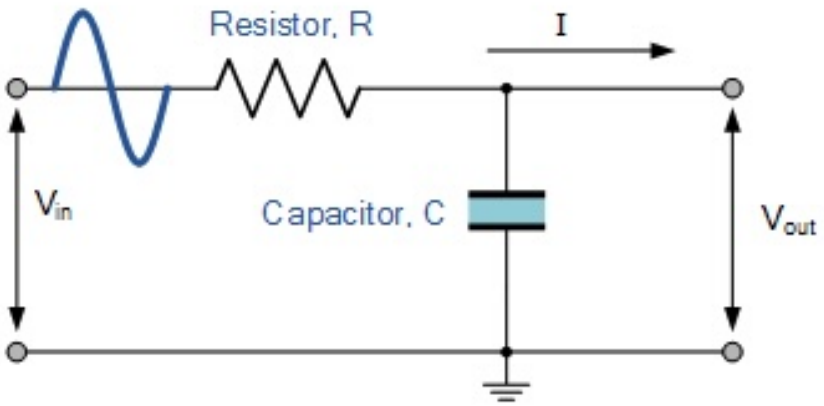
Capacitors Revisited

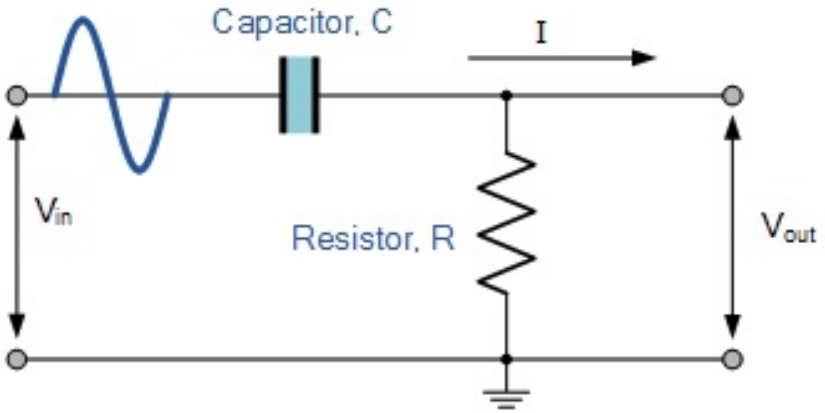


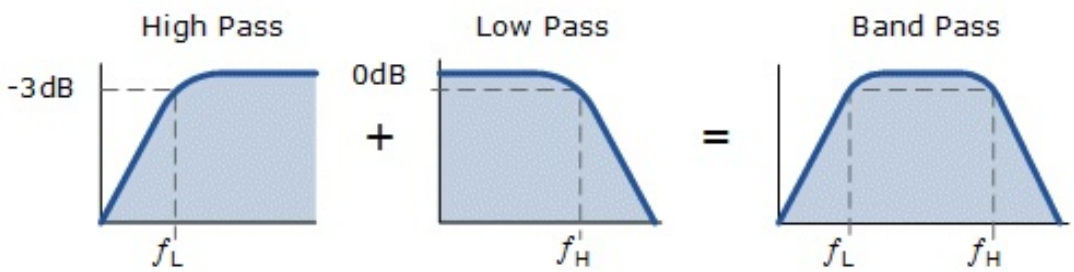
Capacitor acts like:

- High Frequency: Short
- Low Frequency: Open

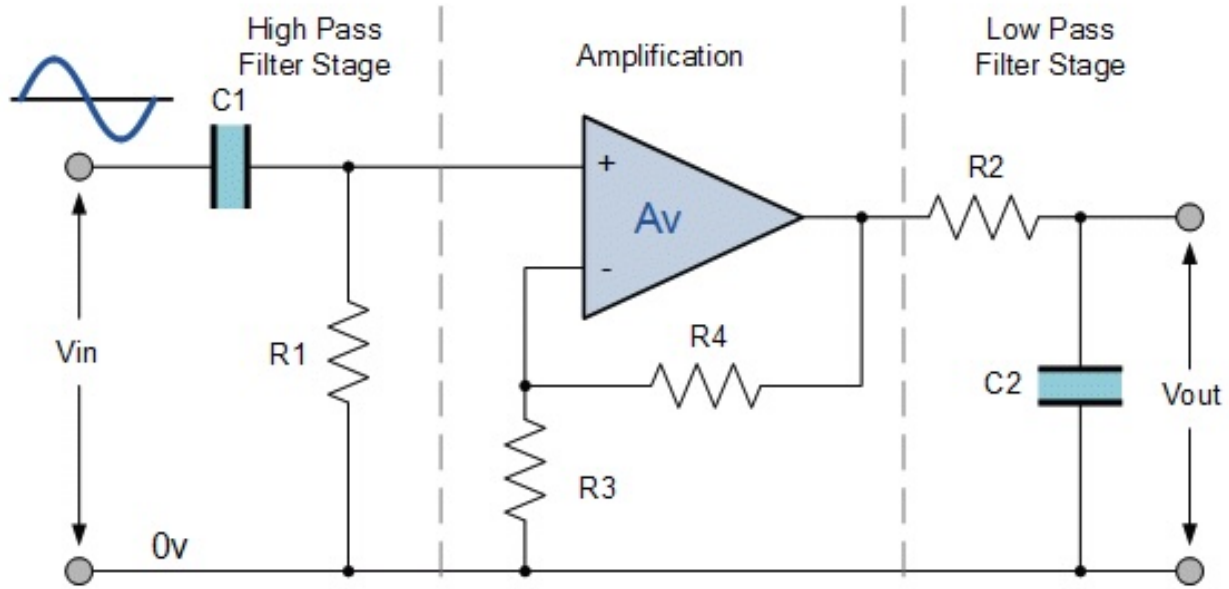
Passive Filters

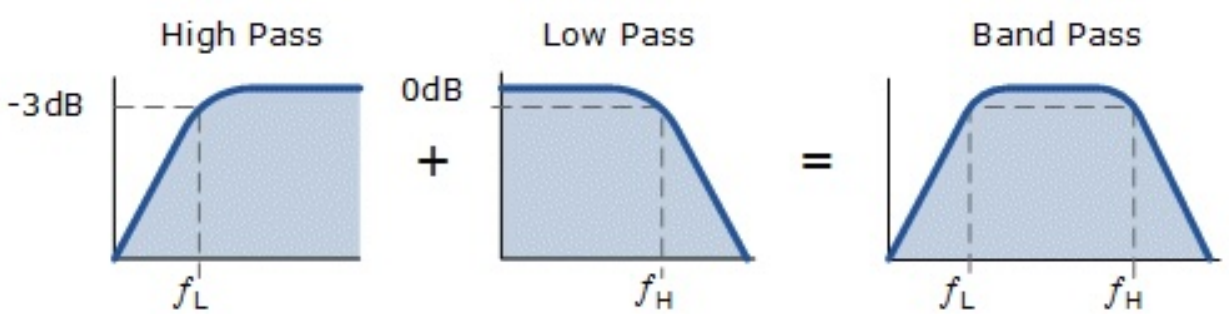




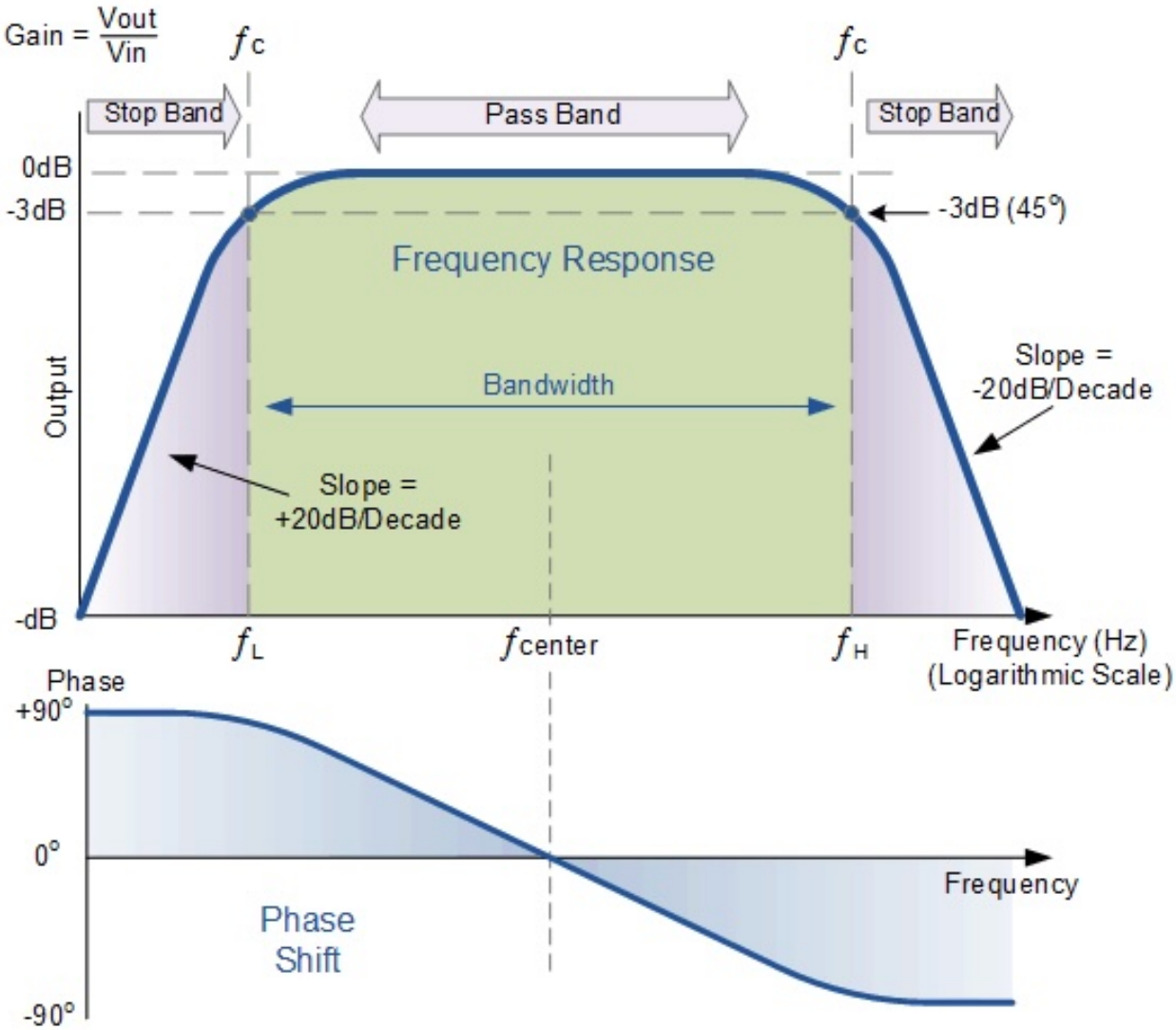


Active Bandpass Filter

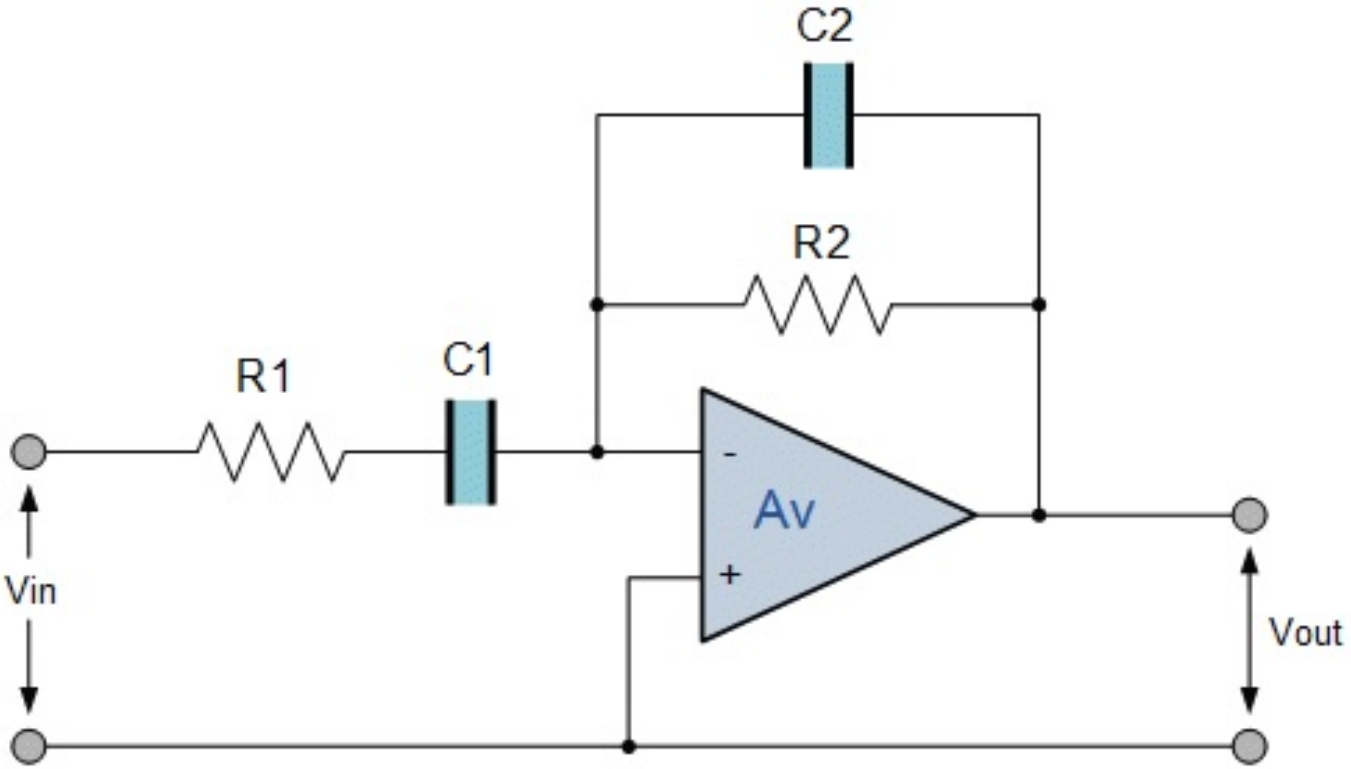




Active Bandpass Response

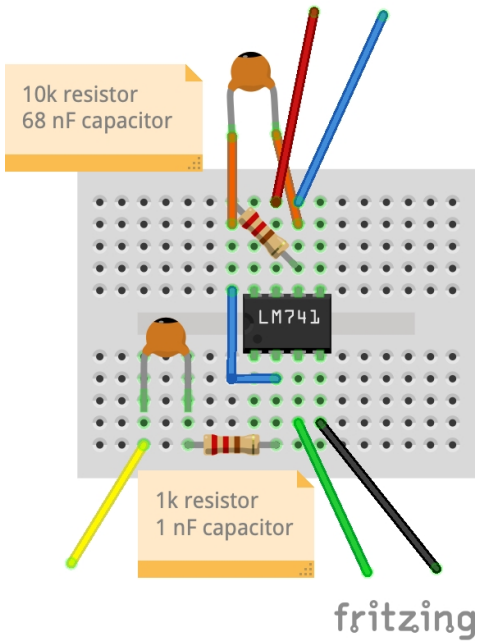


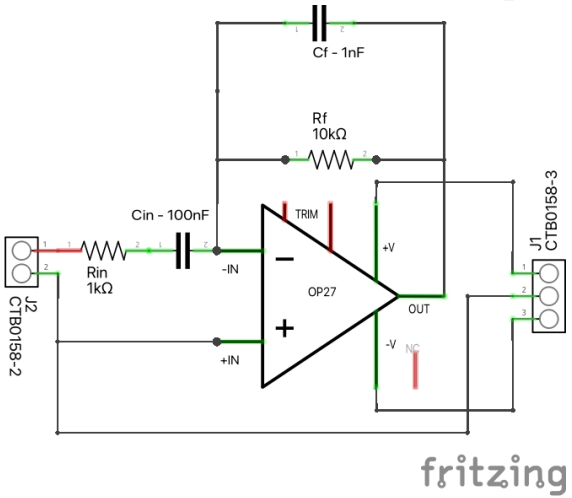
Inverting Active Bandpass Filter



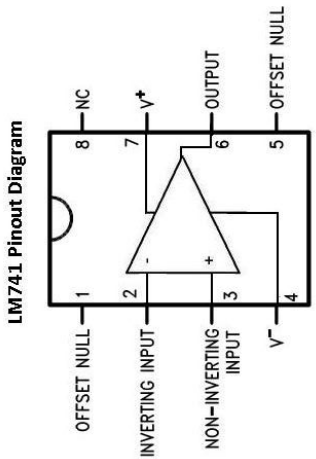
$$f_L = \frac{1}{2\pi R_1 C_1} \qquad f_H = \frac{1}{2\pi R_2 C_2}$$

Assignment: Amplifying BandPass Filter

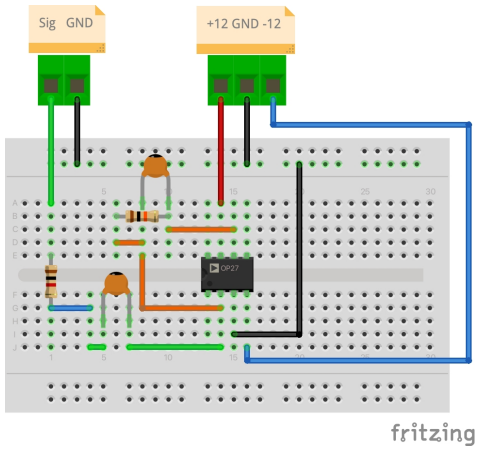


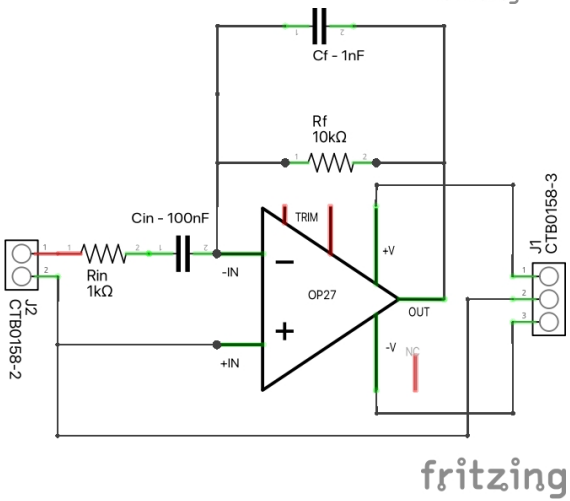


- Start without the C_f capacitor and with shorting the C_{in} capacitor.
- Using a $500mV_{pp}$ input sinewave, you should see $10V_{pp}$ inverted sinewave as the output.
- Install C_f and remove the jumper that is shorting C_{in}.
- Repeat the sweep from the Lowpass Filter Assignment

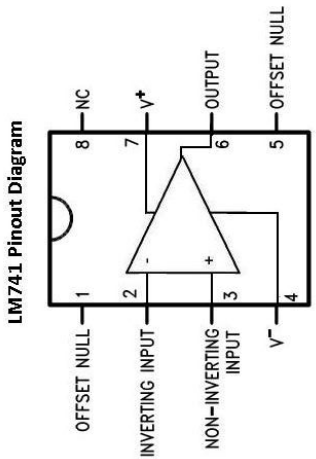


Assignment: Amplifying BandPass Filter

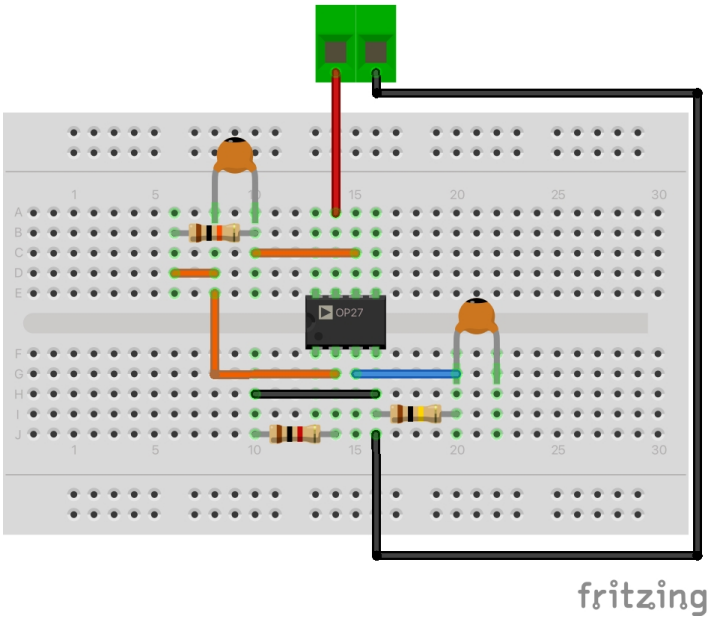


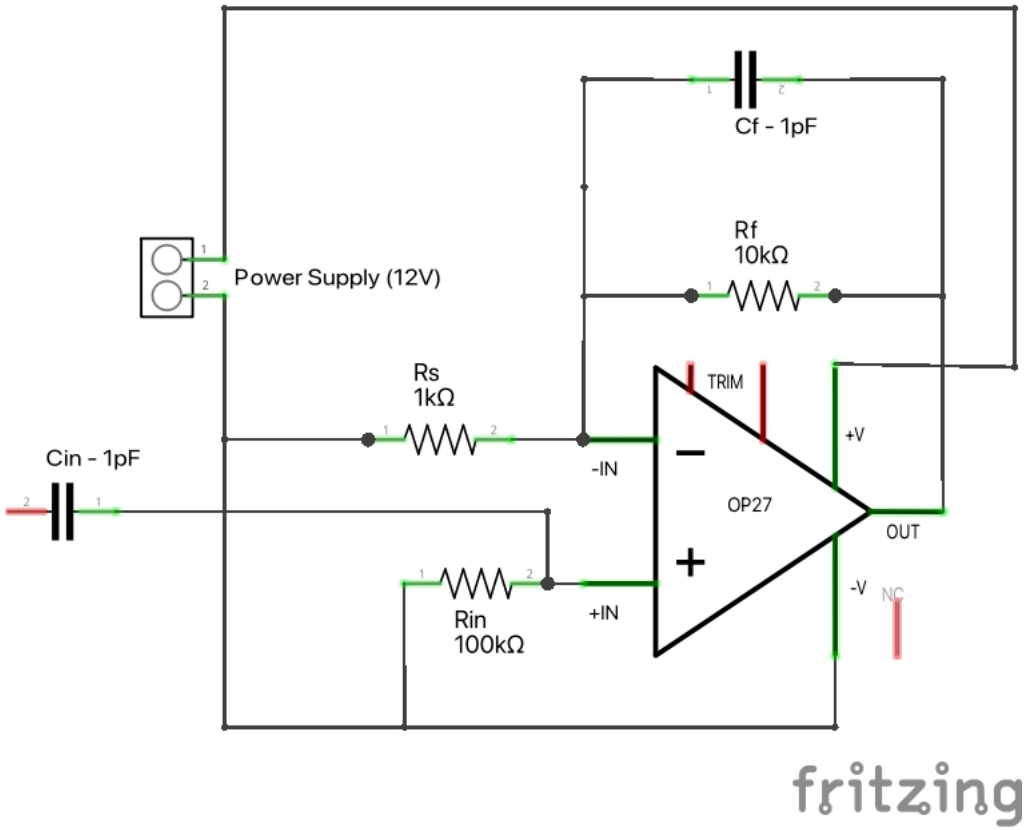


- Start without the C_f capacitor and with shorting the C_{in} capacitor.
- Using a $500mV_{pp}$ input sinewave, you should see $10V_{pp}$ inverted sinewave as the output.
- Install C_f and remove the jumper that is shorting C_{in}.
- Repeat the sweep from the Lowpass Filter Assignment

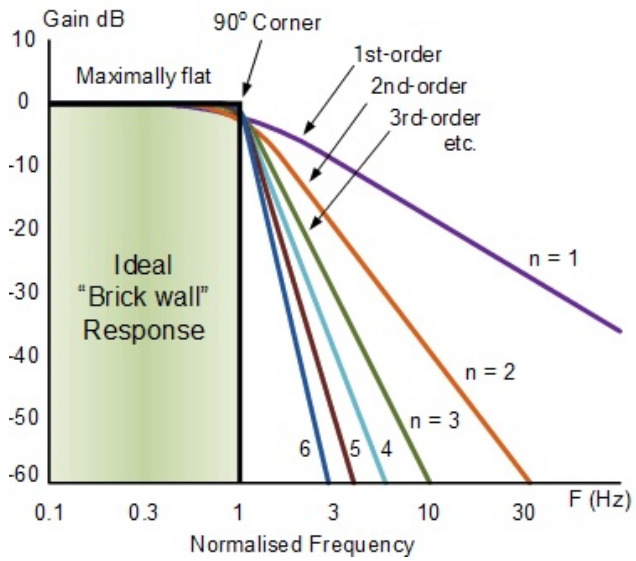


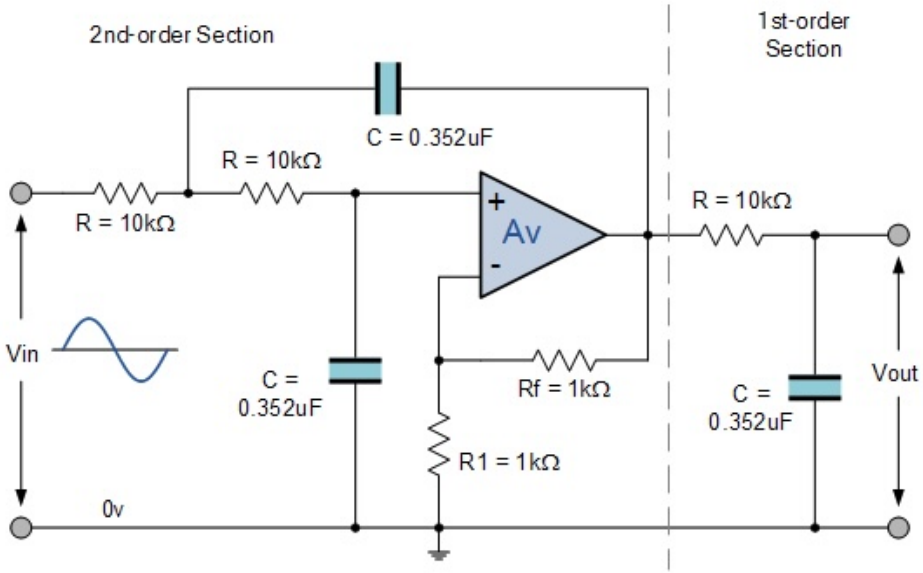
Non-inverting Bandpass filter





Higher Order Butterworth (Lowpass) Filters





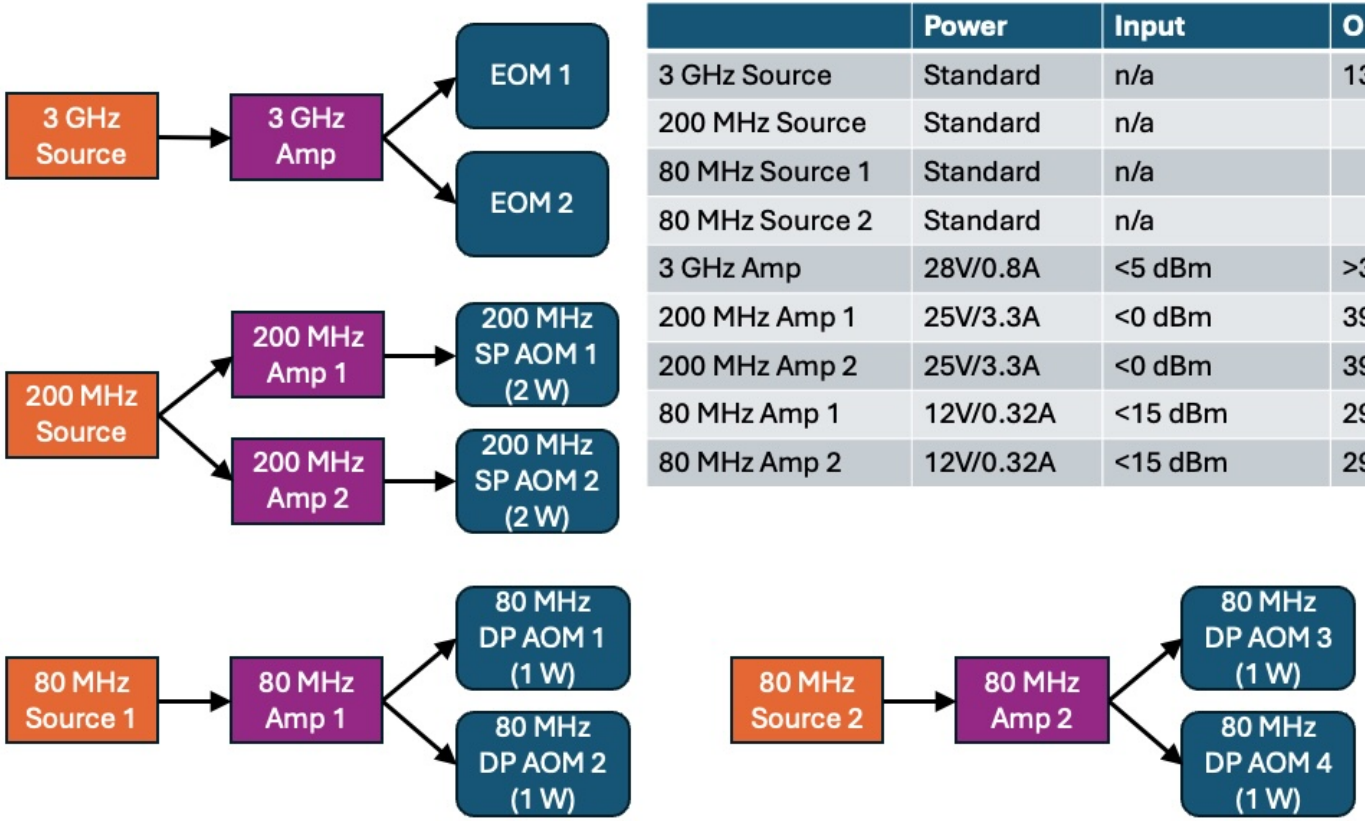
First described in 1930 by the British engineer and physicist Stephen Butterworth in his paper entitled "On the Theory of Filter Amplifiers".

Frequency Generators and Amplifiers

Copyright ©2024 CNM. Frequency Generators and Amplifiers

Frequency and Power Needs

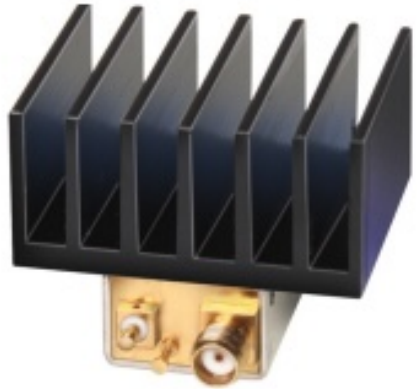
To laser cool, the light needs to be modulated. The modulators will be driven by a signal of a certain frequency and power:



	Power	Input	Output
3 GHz Source	Standard	n/a	13-20 dBm
200 MHz Source	Standard	n/a	
80 MHz Source 1	Standard	n/a	
80 MHz Source 2	Standard	n/a	
3 GHz Amp	28V/0.8A	<5 dBm	>30 dBm
200 MHz Amp 1	25V/3.3A	<0 dBm	39 dBm
200 MHz Amp 2	25V/3.3A	<0 dBm	39 dBm
80 MHz Amp 1	12V/0.32A	<15 dBm	29 dBm
80 MHz Amp 2	12V/0.32A	<15 dBm	29 dBm

Frequency Generators and Amplifiers

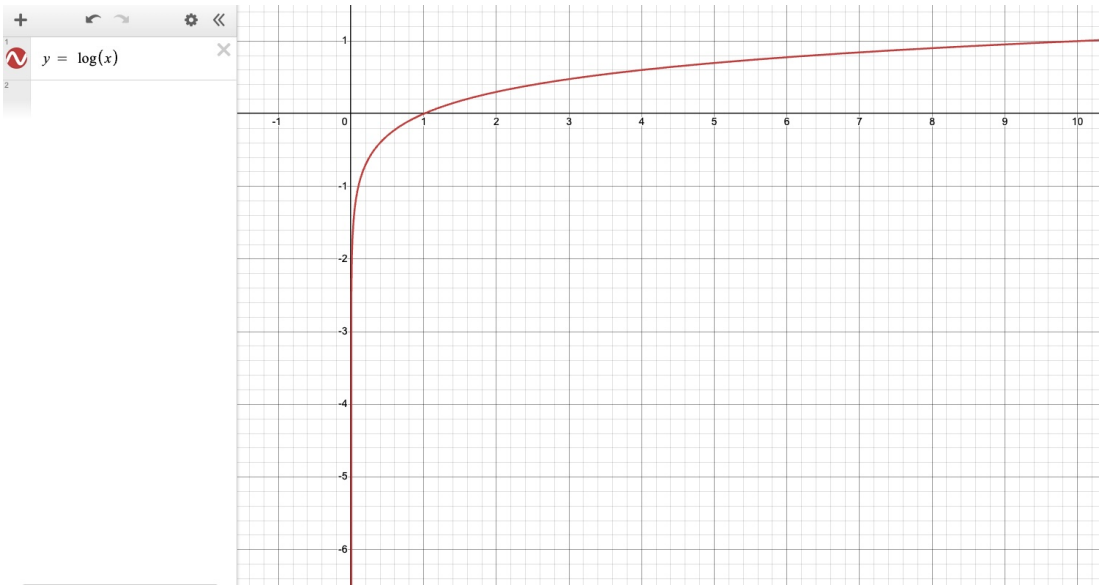






Copyright ©2024 CNM. Frequency Generators and Amplifiers

Recall Logarithms

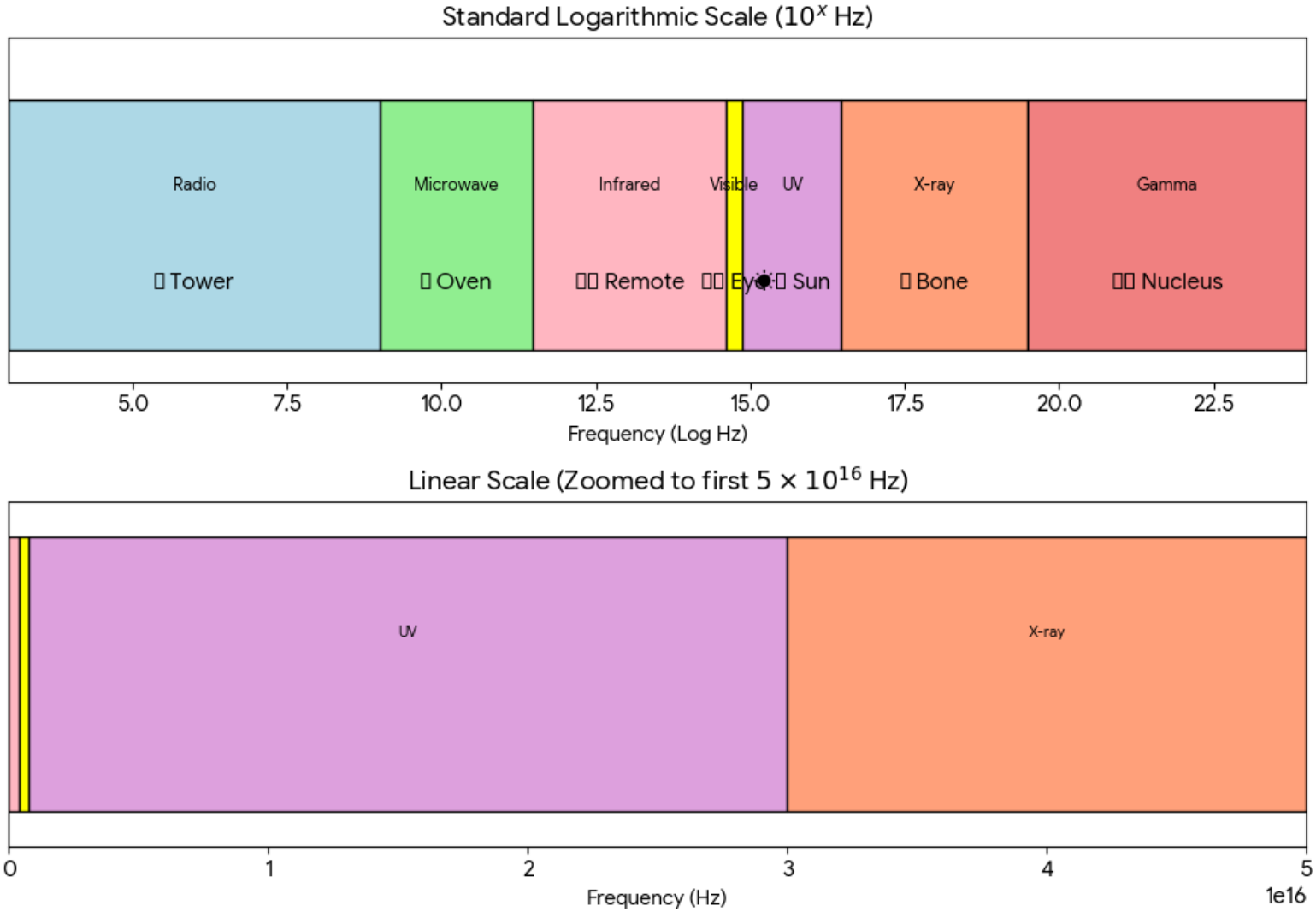


The logarithm of a number (in base 10) is the exponent to which 10 must be raised to produce the number.

- If $x = 10^y$, then $y = log_{10}(x)$
- With logarithms, multiplication becomes addition: $log_{10}(a \cdot b) = log_{10}(a) + log_{10}(b)$

Why Logarithms

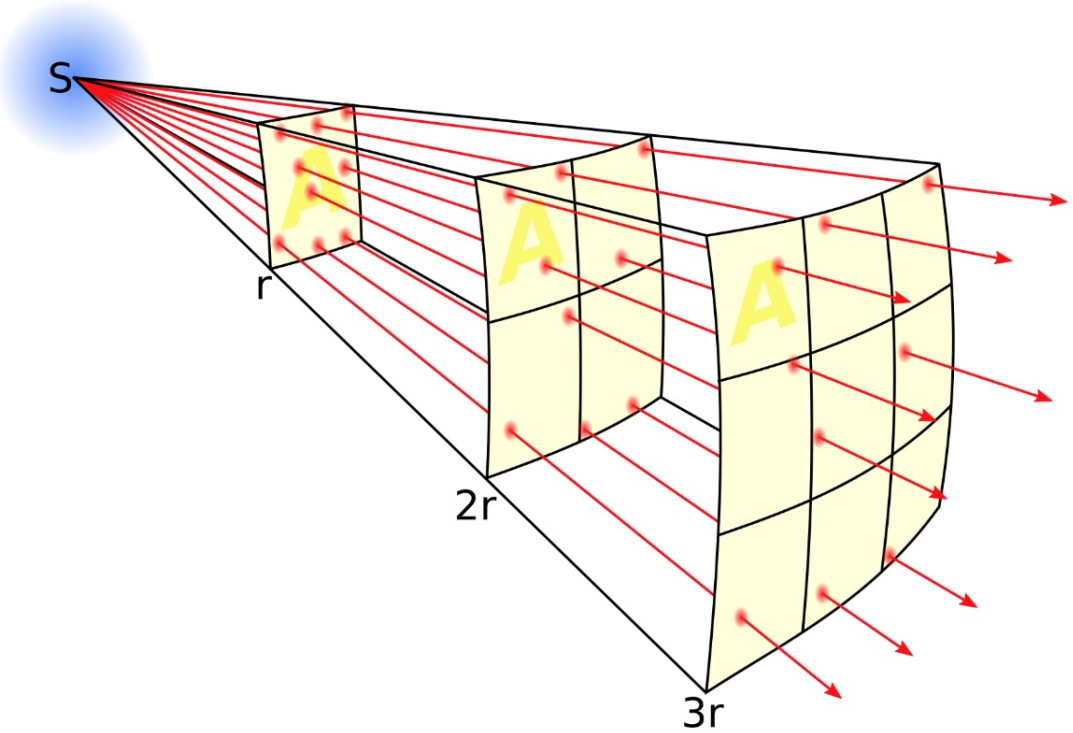
Logarithms are useful when working across many orders of magnitude:



Copyright ©2024 CNM. Frequency Generators and Amplifiers

Why Logarithms

Logarithms started being used in RF communication due to the inverse square law

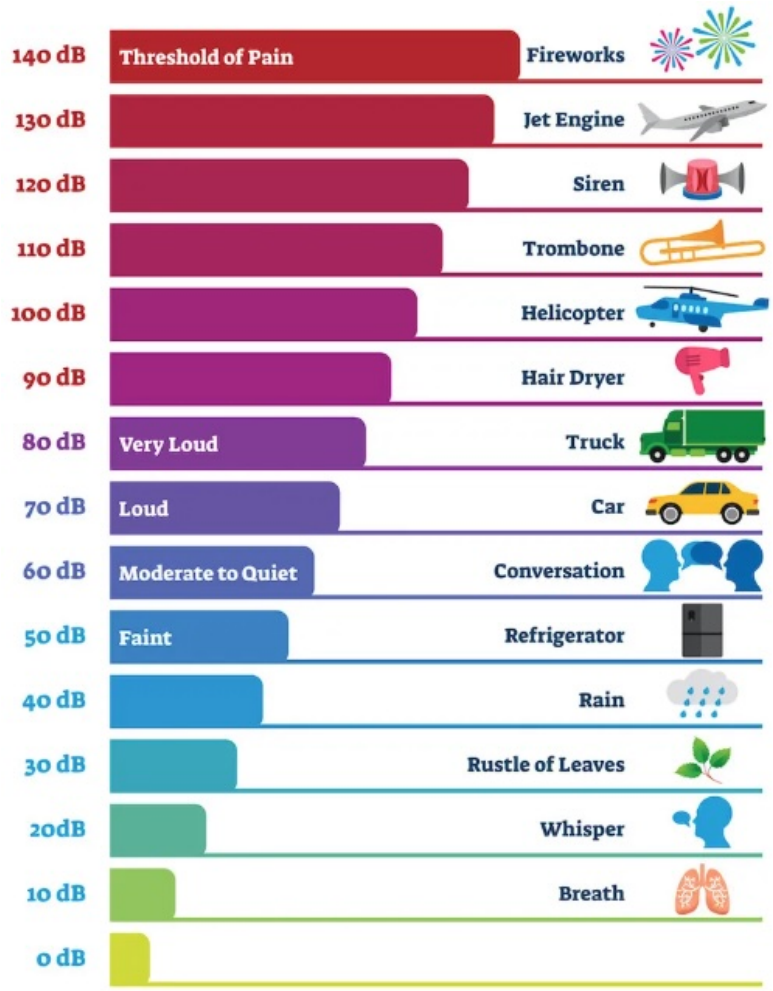


- Surface Area of a Sphere: $A = 4\pi r^2$
- Intensity: $I(r) = \frac{I(1)}{r^2}$

Copyright ©2024 CNM. Frequency Generators and Amplifiers

Decibels (dB)

- A decibel (dB) is an expression of a ratio
- $\text{dB} = 10 * log_{10}(\frac{a}{b})$
- Decibels being a ratio means that they are relative to some reference
- Sound dB (or dB Audio or dBA) expresses the ratio of a sound relative to the threshold of human hearing



Copyright ©2024 CNM. Frequency Generators and Amplifiers

Power Measurements (dBm)

- To measure absolute power in dB, a reference needs to be defined
- It is useful to set the reference to 1 mW:

$$1mW \rightarrow 0dBm$$

- The "m" in dBm indicating the reference is 1 mW.

1 mW	= 0 dBm
2 mW	= 3 dBm
4 mW	= 6 dBm
8 mW	= 9 dBm
10 mW	= 10 dBm
20 mW	= 13 dBm
50 mW	= 17 dBm
100 mW	= 20 dBm
200 mW	= 23 dBm
500 mW	= 27 dBm
1000 mW (1W)	= 30 dBm

Using dB

When using dB, gains and losses (attenuation) are additive.

- Recall: $log_{10}(a \cdot b) = log_{10}(a) + log_{10}(b)$
- Also: $log_{10}(a/b) = log_{10}(a) - log_{10}(b)$
- This leads to: "initial power" + 10 dB = 10 times the initial power
- Examples [1]:
  - $0dBm + 10dB = 10dBm = 10mW$
  - $0dBm + 30dB = 30dBm = 1000mW = 1W$
  - $10dBm + 3dB = 13dBm = 20mW$
  - $20dBm - 3dB = 17dBm = 50mW$
- Another Example: How much is 33 dBm?
  - $33dBm = 33dB$ relative to $1dBm$
  - $33dB = 10dB + 10dB + 10dB + 3dB$
  - $33dBm = 10 * 10 * 10 * 2 * 1mW$
  - $33dBm = 2000mW$ or $2W$

[1] Note that 3dB = doubling

Gooch & Housego 3200-124 AOM

SPECIFICATIONS	
AO Medium	TeO2
Acoustic Velocity	4.2 mm/μs
Active Aperture*	2.5 mm 'L' X 0.32 mm 'H'
Center Frequency (Fc)	200 MHz
RF Bandwidth	50 MHz @ -10 dB Return Loss
Input Impedance	50 Ohms Nominal
VSWR @ Fc	1.3 :1 Max
Wavelength	780-850 nm
Insertion Loss	3 % Max
Reflectivity per Surface	1 % Max
Anti-Reflection Coating	MIL-C-48497
Optical Power Density	250 W/mm^2
Contrast Ratio	1000 :1 Min
Polarization	90 ° To Mounting Plane

PERFORMANCE VS WAVELENGTH	
Wavelength (nm)	**830**
Saturation RF Power (W)	2.0
Bragg Angle (mr)	19.8
Beam Separation (mr)	39.6

PERFORMANCE VS BEAM DIAMETER	
Beam Diameter (μm)	**150**
at Wavelength (nm)	830
Diffraction Efficiency (%)	70*
Rise Time (nsec)	29
Modulation Bandwidth	21.0
Beam Ellipticity	10

Mini-Circuits

COAXIAL

High Power Amplifier

ZHL-5W-1+
ZHL-5W-1X+

50Ω 5 W 5 to 500 MHz SMA Female

FEATURES

- High Power, 5 Watt
- Wideband, 5 to 500 MHz
- High Power Output, +37 dBm Min.
- High Gain, 40 dB Typ.
- Low Noise Figure, 4 dB Typ.
- High IP3, +49 dBm Typ.



Generic photo used for illustration purposes only

Model No.	ZHL-5W-1+	ZHL-5W-1X+*
Case Style	DD0131	
Connectors	SMA female	

+RoHS Compliant
The +Suffix identifies RoHS Compliance.
See our website for methodologies and qualifications

APPLICATIONS

- VHF
- Instrumentation
- Laboratory

ELECTRICAL SPECIFICATIONS[1]

Parameter	ZHL-5W-1+ ZHL-5W-1X+ ▲ Min.	Typ.	Max.	Units
Frequency Range	5		500	MHz
Gain	40	45		dB
Gain Flatness			±1.7	dB
Output Power at 1 dB Compression	+37			dBm
Noise Figure		4.0		dB
Output Third Order Intercept Point		+49		dBm
Input VSWR		2.0		:1
Output VSWR		2.5		:1
DC Supply Voltage		+24	+25	V
Supply Current			3.3	A

1. Specified under matched load to 50Ω.

▲ Heatsink not included. Alternative heat sinking and heat removal must be provided by the user to limit maximum base-plate temperature to +65 °C, in order to ensure proper performance. For reference, this requires thermal resistance of user's external heatsink to be 0.3 °C/W max.

ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature	-20 °C to +65 °C
Storage Temperature	-55 °C to +100 °C
Input RF Power (No Damage)[2]	0 dBm

2. Derate maximum input power by 20 dB when load isn't present.
Permanent damage may occur if any of these limits are exceeded.

Copyright ©2024 CNM. Frequency Generators and Amplifiers

Gooch & Housego 3080-122 AOM

SPECIFICATIONS			
AO Medium			TeO2
Acoustic Velocity			4.2 mm/μs
Active Aperture*	2.5 mm 'L' X	1	mm 'H'
Center Frequency (Fc)			80 MHz
RF Bandwidth	20 MHz @	-10 dB	Return Loss
Input Impedance			50 Ohms Nominal
VSWR @ Fc			1.3 :1 Max
Wavelength			780-850 nm
Insertion Loss			3 % Max
Reflectivity per Surface			0.25 % Max
Anti-Reflection Coating			MIL-C-48497
Optical Power Density			250 W/mm^2
Contrast Ratio			1000 :1 Min
Polarization			90 ° To Mounting Plane

PERFORMANCE VS WAVELENGTH	
Wavelength (nm)	**830**
Saturation RF Power (W)	1
Bragg Angle (mr)	7.9
Beam Separation (mr)	15.8

PERFORMANCE VS BEAM DIAMETER			
Beam Diameter (μm)	**200**	**250**	**500**
at Wavelength (nm)	830	830	830
Diffraction Efficiency (%)	70	80	85
Rise Time (nsec)	34	41	80
Modulation Bandwidth	15.9	12.65	6.3
	15	10	1

Mini-Circuits

COAXIAL

Medium High Power Amplifier

ZX60-100VH+
ZX60-100VHX+

50Ω 0.3 to 100 MHz

ELECTRICAL SPECIFICATIONS AT +25°C

Parameter	Condition (MHz)	ZX60-100VH+ ZX60-100VHX+** Min.	Typ.	Max.	Units
Frequency Range		0.3		100	MHz
Gain	0.3 - 100	33	36	--	dB
Gain Flatness	0.3 - 100	--	±0.3	--	dB
Output Power at 1dB Compression	0.3 - 100	--	+30	--	dBm
Output third order intercept point	0.3 - 100	--	+43	--	dBm
Noise Figure	0.3 - 100	--	4	--	dB
Input VSWR	0.3 - 100	--	1.6	--	:1
Output VSWR	0.3 - 100	--	1.5	--	:1
Active Directivity (Isolation-Gain)	0.3 - 100	--	14	--	dB
DC Supply Voltage		--	+12*	--	V
Supply Current		--	320	370	mA

*Recommended Operating Voltage.

**Heat sink not included. Alternative heat sinking and heat removal must be provided by the user to limit maximum base-plate temperature to 85°C, in order to ensure proper performance. For reference, this requires thermal resistance of user's external heat sink to be 3.3°C/W max.

ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature (ground lead)	-40°C to 85°C
Storage Temperature	-55°C to 100°C
Power Dissipation	4.4 W
Input RF Power (no damage)	+15 dBm
DC Voltage	+13V

Permanent damage may occur if any of these limits are exceeded.

Copyright ©2024 CNM. Frequency Generators and Amplifiers

MOT Modulation Scheme

